

OrthoHMM publication working draft

OrthoHMM: HMM-Centered Orthogroup Inference And Phylogenetic Refinement

Working manuscript, updated 25 September 2026. Not submission-ready. Sections below distinguish completed development-exposed analyses from prospective work. Internal evidence links are supplied for audit. The [selected bibliography](#) contains 37 records. Its [provenance and review history](#) are retained separately; citation coverage and journal-specific formatting remain incomplete.

Study Objective

We evaluate an HMM-centered approach to orthogroup inference and its integration with phylogenetic refinement. The principal questions are whether profile-based expansion contributes measurable recovery beyond the initial search, whether broader candidate groups improve reconciliation, and how accuracy and computational costs compare with established tools. These questions are not equivalent to asking whether a single aggregate score is maximized. We distinguish homolog-group recovery, resolved ortholog-pair prediction, and conserved-family recovery throughout the study.

Methods

Configurations And Comparator Outputs

The retained OrthoHMM configurations are high sensitivity and the `satellite_v2` phylogenetic pipeline. The prospective validation configuration is pinned to source revision `7f3a9e4`, with BLOSUM62, E-value threshold `1e-4`, Leiden CPM resolution `0.1`, seed `4`, and the default refinement profile. The Leiden optimizer is described by [Traag et al. \(2019\)](#) and the CPM objective by [Traag et al. \(2011\)](#). These clustering references do not imply biological orthology guarantees. The graph library is cited separately as [Csardi and Nepusz \(2006\)](#), `igraph`; runtime manifests, not this citation, identify executed library versions. The phylogenetic configuration uses `satellite_v2` candidate expansion, an internally inferred species tree, minimum-variance species-tree rooting, the species-overlap root rule, and positive-paralogy pair inference. Historical runs retain their actual source and configuration records; the prospective pin must not be retroactively attributed to them. See the [frozen validation protocol](#). The original OrthoHMM lineage is cited as

the [Steenwyk et al. \(2024\) preprint](#); the implementation examined here includes later changes that require the source/runtime description rather than attribution to that preprint alone.

The [method diagram](#) distinguishes initial HMM search, cluster-profile expansion, high-sensitivity orthogroups, candidate-family construction, gene/species-tree inference and constraint-guided membership decisions. Root HOGs and inferred ortholog pairs are separate output levels. This schematic describes the intended frozen workflow, not evidence of execution correctness or an HMM accuracy advantage.

Comparators comprise OrthoFinder 3.1.5 full and its sequence-only MCL checkpoint, OrthoMCL 1.4, SonicParanoid 2.0.9, ProteinOrtho 6.3.6, and FastOMA 0.3.5. The checkpoint is a distinct output level, not another phylogenetically finalized prediction. In the retained OrthoFinder full run, finalized `Orthogroups.txt` reflects root hierarchical-group postprocessing; its filename does not identify it as sequence-only. FastOMA's supplied OrthoFinder tree is disclosed rather than treated as independent tree inference. Native-output semantics and source records are retained in the [comparison](#) and [OrthoBench protocol](#). The v3 method is described by [Emms et al. \(2026\)](#), with a [published Figure2 correction](#). This citation does not replace executable/version provenance for our runs. The earlier orthogroup and phylogenetic methods are described by [Emms and Kelly \(2015\)](#) and [Emms and Kelly \(2019\)](#). Other comparator references are [Li et al. \(2003\)](#), [OrthoMCL](#), [Cosentino et al. \(2024\)](#), [SonicParanoid2](#), [Klemm et al. \(2023\)](#), [Proteinortho6](#), and [Majidian et al. \(2025\)](#), [FastOMA](#). These method citations do not establish which optional modules or settings ran in our experiments; retained execution records govern that attribution.

Companion-method references distinguish protein search ([DIAMOND](#)), multiple alignment ([MAFFT](#) and [FAMSA](#)), tree inference ([FastTree](#) and [FastME](#)), and protein-family clustering ([MCL](#)). These references describe method lineages, not proof that every installed companion was invoked in every run. Version resolution and actual commands are retained separately in the [dependency review](#) and [DGX companion-path audit](#).

For QfO, OrthoHMM high sensitivity and the OrthoFinder checkpoint contribute group-derived cross-species pairs; OrthoHMM satellite_v2 and full OrthoFinder contribute phylogenetically inferred pairs. SonicParanoid contributes native species-pair relations, ProteinOrtho its native post-clustering graph, and FastOMA native pairs. OrthoMCL contributes cross-species pairs derived from its final MCL groups. Pre-MCL graph edges are retained only as a diagnostic. These output differences are part of the practical pipeline comparison and must not be interpreted as a matched search-engine ablation.

A complete historical FastOMA conversion audit validated 15,320,615 native cross-species pairs against the 976,504-protein input universe and found no duplicates. Independent sorting reproduced both the retained raw prediction set and the 15,277,489-pair QfO-filtered set exactly; 43,126 pairs contained identifiers absent from the historical mapping. This establishes conversion integrity, not biological correctness or freedom from mapping-related bias, and does not substitute for the corrected-release rerun. [Conversion audit](#).

The historical FastOMA OrthoXML declared 967,184 of the 976,504 input proteins. Its 55,486 root HOGs referenced 588,873 proteins, of which 584,445 appeared in native pairs. The root-HOG table exactly matched XML membership, and all native pairs joined known cross-species proteins within one root HOG. These are native-output coverage counts, not reference-relative recall. Identifier consistency does not establish why proteins were omitted or whether each predicted relation is biologically correct. [OrthoXML audit](#).

Development-Exposed Benchmarks

QfO and OrthoBench influenced development and are therefore not independent confirmation sets. OrthoBench evaluates the complete retained 70-RefOG panel with its low-certainty exclusions and audited weighted pair statistic. The benchmark frameworks are described by [Altenhoff et al. \(2016\)](#) and [Emms and Kelly \(2020\)](#), respectively. QfO service updates are described by [Altenhoff et al. \(2020\)](#) and [Nevers et al. \(2022\)](#), including the 2020 reference-proteome update and VGNC benchmark in the latter. These papers do not replace retained release, input and scoring provenance. Weighted counts divide each reference-family contribution by reference size minus one; aggregate precision, recall, and F1 are computed from those counts, not by averaging family F1 values.

We resampled RefOGs jointly across methods 20,000 times using NumPy PCG64 seed 20260916 and recomputed the statistic for each replicate. We report paired percentile intervals and Bonferroni-adjusted intervals across two OrthoHMM-versus-OrthoFinder contrasts and three metrics. The protocol was recorded after historical aggregate results were known, before this bootstrap; it is not preregistration of method selection. Family exchangeability and residual cross-family dependence limit inference. [Protocol](#).

The numerical and reporting implementations use [NumPy](#) for array/bootstrap operations, [Biopython](#) for sequence/tree validation helpers, and [Matplotlib](#) for plots. The [analysis-dependency review](#) links those roles to source modules; per-run manifests determine executed versions. Library citations do not validate statistical assumptions or plots.

The [SwissTree resource](#) and [TreeFam resource paper](#) describe curated gene-tree references; they do not authenticate our exact retained snapshots. [FAS](#) measures feature-architecture similarity, not pairwise orthology F1. Its later method paper is resource attribution, not proof of the implementation used for the QfO2020 archive. The [ENZYME resource paper](#) provides enzyme-nomenclature background, not a citation for the QfO EC scoring formula or evidence of the source release used for our historical annotations. Source checks and machine-readable SwissTree/ENZYME citations are retained in the [resource supplement](#).

The [VGNC resource paper](#) describes vertebrate gene nomenclature, while the QfO service protocol defines its benchmark use. Gene Ontology is attributed to its [foundational paper](#); the later [2026 knowledgebase update](#) is contextual resource attribution, not the annotation release or scorer used in QfO2020. Neither resource citation establishes historical snapshot identity or supplies dependency-aware uncertainty for our functional endpoints.

For QfO we retain individual VGNC, SwissTrees, TreeFam-A, EC, GO, and FAS endpoints, their native axes, and source provenance. The unweighted mean of six project-selected summaries is secondary and is not an official QfO score. Relation counts assessed by functional endpoints do not necessarily equal total prediction coverage. Three Kingdoms is supplementary: its BUSCO-reference pair statistic ignores false positives involving non-reference genes and cannot establish proteome-wide orthology accuracy. BUSCO's conserved-gene completeness framework ([Manni et al., 2021](#)) is distinct from this project's BUSCO-reference pair statistic. [Machine-readable comparison](#).

The [retained Three Kingdoms source audit](#) identifies BUSCO5.8.2 and eukaryota_odb10 dated 2024-01-08 (OrthoDB10.1). Its 12 saved full tables reconstruct the scored reference byte-for-byte: 255groups, 2,035genes and 7,352pairs. Compressed downloads reproduce retained raw FASTAs; eleven staged proteomes are byte-identical, while seven zebrafish

sequences differ only by removal of 29 stop markers. None of those seven belongs to the scored reference; indirect effects and historical per-tool input identity are not established by that observation. The Xenopus download URL uses a moving UniProt current_release path, so retained hashes are needed and the URL alone is not a reproducible release identifier.

A subsequent method-input audit found that SonicParanoid's native snapshot matches the raw zebrafish FASTA, not the stop-marker-stripped staged version. Five other retained checksum manifests match all 12 staged inputs; the older high-sensitivity metrics lack per-file hashes. The historical supplementary table is therefore not presented as a fully established identical-input cohort. A contemporary matched-staged-input Sonic run has now completed with independently verified conversion and scoring. The updated supplementary table uses this new run and retains the historical Sonic result as a separate diagnostic row. No causal effect of the input mismatch is inferred from the difference between runs; other historical input-consumption limitations remain unresolved.

For the separately recovered four-stage QfO replay, SwissTrees raw counts were reconstructed using the frozen native scorer and reference. Each one-direction reference relation contributes half a count before a prior of one is added to each confusion category. Across 18 reference families, precision and recall are averaged separately before forming the project F1. We used 100,000 shared paired family draws (PCG64seed20260919), recomputing this statistic in each draw, with nominal 95% and 12-endpoint Bonferroni percentile intervals for four contrasts and three metrics. The protocol was fixed after point estimates were known but before interval calculation; it is development-exposed follow-up, not independent confirmation. Disjoint represented genes do not eliminate dependence from shared evolutionary history or merged predictions. Other QfO challenges need separate uncertainty analyses; these intervals do not cover the secondary six-metric mean. Protocol.

Prospective Validation And Ablations

YGOB combines curated homology with syntenic context (Byrne and Wolfe, 2005); that original resource description does not establish independence of our retained subset.

The frozen YGOB experiment retains 16 non-Saccharomyces species, 83,404 inference proteins, and 83,391 reference genes in 10,250 curated pillars. Thirteen proteins from ambiguous duplicate-membership rows remain in inference but are projected out of scoring. The endpoint is homolog-group co-membership, including within-species pairs, not resolved orthology. The primary contrast is satellite_v2 versus full OrthoFinder; high sensitivity is secondary and the sequence checkpoint diagnostic. Its 20,000-replicate paired pillar bootstrap has seed 20260917 and six-metric multiplicity handling. Native/source/input, command/conversion, independent reference reconstruction and overlap checks passed before scoring. Direct pair enumeration independently reproduced all four methods' TP/FP/FN counts. Frozen protocol.

On this frozen panel, satellite_v2 achieved F1 92.233654%, versus 92.318524% for full OrthoFinder. The paired difference was -0.084870 percentage points (nominal 95% interval [-0.487503, 0.312676]; six-endpoint Bonferroni interval [-0.622528, 0.445222]). This does not establish superiority or equivalence. Satellite_v2 had higher precision (+6.401035 points) and lower recall (-6.914033 points), with both adjusted intervals excluding zero. High sensitivity scored 82.038608% F1, below full OrthoFinder by 10.279916 points (adjusted interval [-11.260924, -9.298064]). The diagnostic sequence-

only OrthoFinder checkpoint scored 85.572844%. All methods covered every reference gene. No configurations were changed in response to these held-out outcomes. Frozen results, machine-readable evidence.

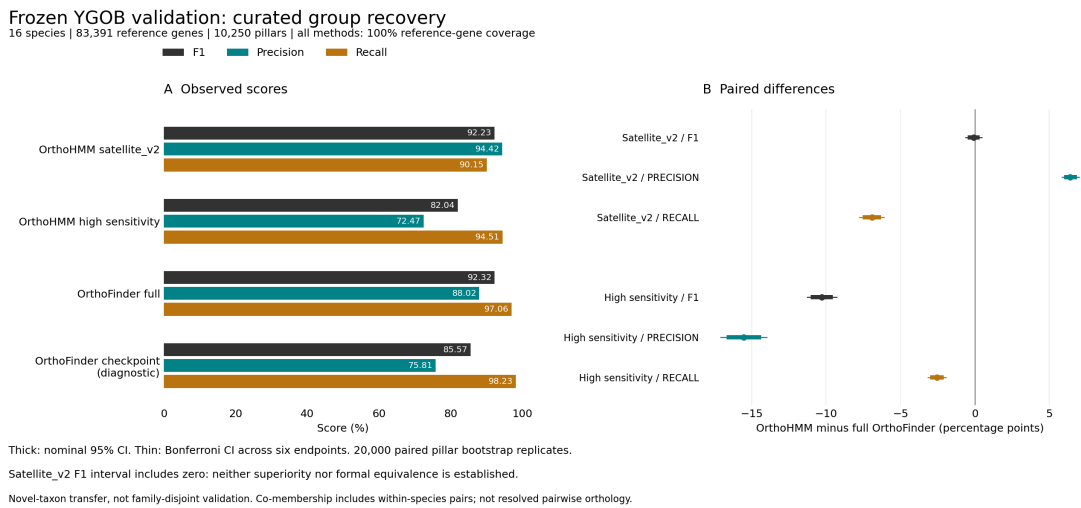


Figure: observed co-membership scores and paired OrthoHMM-minus-OrthoFinder differences. Thick intervals are nominal 95%; thin intervals apply the frozen six-endpoint Bonferroni adjustment. The checkpoint is diagnostic. All methods cover every scored reference gene; this does not establish family independence.

This experiment tests novel-taxon transfer, not family-disjoint validation. The completed label-independent homology screen found qualifying development input hits for 71,714 of 83,404 proteins and 6,952 of 10,250 pillars. Absence of a qualifying hit does not establish absence of homology. Shared reference-resource limitations are documented separately. Screen, resource audit.

Prospective component experiments cross profile-HMM expansion, candidate expansion, and reconciliation in eight cells. Expansion-plus-reconciliation cells retain their own membership constraints; independently inferred trees are distinguished from supplied-tree diagnostics. A profile-expansion-off arm still uses the HMM-centered initial search. A matched sequence-search control is required before attributing an overall advantage to HMMs. The OrthoBench, original-release QfO and corrected-release QfO factorials are complete. Each QfO factorial has paired SwissTrees intervals across 18 families and 42 adjusted endpoints. The corrected full-comparator panel and better-matched search sensitivity or effort remain unfinished. Original-release results are retained as development-exposed evidence, not relabeled as corrected-release validation. Ablation protocol.

Evolutionary Simulations And Runtime Admission

We used Zombi at revision 8db13ee4ba007f46c17f38586d31e5aa617c1647 and Pyvolve 1.1.0 with the recorded per-family seed adapter. These published simulator methods do not substitute for local checks of event-derived truth and deterministic sequence generation.

Two separate panels each contain ten independent simulation seeds and seven conditions: baseline, increased divergence, increased duplication/loss, combined divergence and turnover, missing proteins, uneven taxon sampling, and a taxon-count control. Native Zombi histories provide event-based cross-species orthology truth; shared ancestral-

family membership alone is not treated as resolved orthology. Four configurations are evaluated on each of the 70 datasets. The first panel uses fixed 300-residue sequences; a prospectively specified second panel assigns each ancestral family a deterministic seed-specific length between 100 and 500 residues. Descendants retain that length, so this panel does not simulate within-family indels. [Fixed protocol](#), [variable-length protocol](#).

Initial OrthoHMM runs lacked a required native alignment library; profile construction exceptions were silently caught and no profiles were built. Those runs are invalid runtime diagnostics, not estimates of the intended method. Corrected runs retain the frozen algorithm and verify compiled libraries and an actual profile-construction probe before and after execution. Valid original comparator outputs are reused with separate provenance. Admission requires native completion and output validation, including finite OrthoFinder graph weights; a zero process exit code alone is insufficient. [Runtime correction](#).

Within each panel, we bootstrap paired successful seeds 20,000 times and recompute the mean seed-level statistic. Fourteen primary F1 contrasts use Bonferroni-adjusted intervals; precision and recall are exploratory. Failed runs are reported separately, not assigned zero accuracy. Complete-case contrasts condition on joint success, and available-case table means cannot be subtracted when they summarize different seeds. The two panels are not pooled. [Fixed results](#), [variable results](#).

Simulation Species-Tree Controls

For each of the 70 variable-length datasets, the frozen `satellite_v2` and full OrthoFinder pipelines were rerun with three prespecified supplied trees: the induced generating topology and deterministic rooted NNI perturbations at clade distances 2 and 4. Branch lengths travel with subtrees; these are topology stress tests, not sampled posterior uncertainty. Generating trees are oracle diagnostics, not independently inferred predictions or guaranteed accuracy upper bounds. Non-tree settings and input sequences remain frozen. All 132 available unchanged-inferred-tree mode controls reproduced native pairs and retained upstream files; 199 downstream partition checks also agreed. Eight unavailable original controls were retained, not excluded from subsequent runs.

Independent admission checked all 420 supplied runs, requested topology retention, native output validity and source/input provenance. Cross-arm retained-artifact comparisons preceded truth scoring. All successful original inferred scores were recomputed exactly. For each condition and method we compared generating minus inferred, NNI1 minus generating, and NNI2 minus generating for F1, precision and recall. All 126 endpoints are exploratory, with 20,000 paired-seed resamples and Bonferroni tail adjustment at family $\alpha 0.05$. The statistic is the mean paired per-seed difference, not a pooled pair-count ratio. Failed arms remain explicit without accuracy imputation. Small seed counts and bootstrap-tail resolution limit interval calibration. [Protocol](#), [complete results](#).

Results

OrthoBench Shows A Precision-Recall Tradeoff

OrthoHMM `satellite_v2` achieved weighted F1 74.106074%, compared with 72.736480% for full OrthoFinder and 70.358998% for high sensitivity. `Satellite_v2` precision was 81.770454% and recall 67.755336%; full OrthoFinder precision was 66.065103% and

recall 80.906577%. The satellite_v2 F1 difference was +1.370 percentage points, with nominal 95% interval [-4.504, 7.917] and adjusted interval [-6.290, 10.627]. These intervals do not establish an F1 advantage. Its precision difference was +15.705 points and recall difference -13.151 points; corresponding adjusted intervals excluded zero in opposite directions. These are development-exposed comparisons, not selection-adjusted confirmation. [Uncertainty results](#).

Across individual RefOGs, satellite_v2 had 23 F1 wins, 10 ties, and 37 losses relative to full OrthoFinder. High sensitivity had 20 wins, 11 ties, and 39 losses. These descriptive counts need not rank methods identically to the weighted aggregate statistic and do not substitute for it. [Per-family analysis](#).

OrthoBench Direct Hits And Grouping Outcomes

The [joint search/stage trace](#) covers all 70 retained reference families. Of 40733 descriptive pair memberships, 24333 lack a direct cached hit in either direction. Final root HOGs nevertheless group 8465 of these together, while 1971 pairs with at least one retained hit remain separated. Candidate-to-root processing separates 1575 previously grouped reference pairs and gains none. These observations show why direct-hit presence alone cannot explain final membership. They do not identify rejected RBNH edges, prefilter-versus-scoring rejection, or the individual effects of tree inference, reconciliation and constraints. Counts include low-certainty and within-species pairs and are not the official weighted recall statistic or independent biological validation.

A subsequent [frozen initial-graph reconstruction](#) separates retained hits from accepted graph edges. Among the 1,971 hit-supported pair memberships separated in final root groups, 505 had an initial edge, 1,453 had hit scores below both endpoint thresholds, and 13 had no finite endpoint threshold. Conversely, 4,201 hit-supported memberships without an initial edge were grouped together in final root groups. These descriptive counts passed an independent pair-table and arithmetic audit, but the audit does not independently reconstruct the native graph. Initial direct edges are neither necessary nor sufficient for final co-membership; indirect paths, subsequent refinement and partitioning require separate causal investigation.

A separately frozen CPU [search-decision diagnostic](#) observed all 81,466 directed reference-family pairs across 144 species directions. It accepted 31,479 pairs, excluded 48,368 at the prefilter and scored 1,619 without reaching $E < 1e-4$. Accepted-hit presence matched the historical trace for every watched pair. Independent reconstruction from saved raw candidate arrays verified every decision and table row. This localizes rejection in the observed run but does not authenticate historical execution or prove numerical score equivalence. The descriptive pair set includes within-species and low-certainty members; these counts are not official recall, evidence that excluded pairs are true orthologs, or a causal explanation of final grouping errors. Excluded pairs were not scored counterfactually, and benchmark predictions were not changed.

Joining these decisions to the retained final-root-group trace found 8,386 pairs excluded by the prefilter in both directions that were nonetheless grouped together, and 15,129 such pairs that remained separated. Pairs scored but not significant in both directions contributed 32 grouped and 218 separated observations; mixed prefilter/scoring rejection contributed 47 and 521. These recover the earlier 8,465 grouped and 15,868 separated pairs without any accepted direct hit. The [complete family-level join](#) also retains the 1,971 separated pairs with an accepted direct hit. These descriptive patterns reinforce that direct search acceptance is neither necessary nor sufficient for final co-membership, not that prefilter exclusions are harmless or causally responsible for the observed F1 deficit.

QfO Results Vary Across Endpoints

The secondary QfO means were 0.782071 for full OrthoFinder, 0.748243 for satellite_v2, and 0.682548 for high sensitivity. Full OrthoFinder had higher VGNC, SwissTrees, and TreeFam-A F summaries than either OrthoHMM mode. Satellite_v2 had higher EC, GO, and FAS similarity summaries than full OrthoFinder. Because these functional summaries assess different predicted relation sets, higher similarity alone does not establish higher overall orthology accuracy. The six native endpoint panels retain their assessed relation counts or precision-recall axes. [Comparison](#), [endpoint figure](#).

An audit of saved FAS samples reproduced all eight comparator means and native standard errors, as well as four recovered-stage results, within 1e-12. The scored fraction of reported eligible pairs ranged from 0.0067% to 31.3477%; the eligible count is not the total prediction count. Native code subsamples without setting a shuffle seed, and its pair-based SEM does not account for shared proteins or families. These SEMs are not dependence-aware uncertainty for method contrasts. The audit validates saved-sample arithmetic, not sampling representativeness or underlying feature-score correctness. [FAS sample audit](#).

For GO and EC, all 24 historical-comparator/recovered-stage raw counts, means and native uncertainty values were consistent with the retained scorers within six-decimal raw serialization bounds. Inspection and execution of the pinned Darwin statistics implementation established that GO/EC stderr values are Student-t 95% confidence half-widths, whereas FAS reports one SEM. These native quantities are not interchangeable and neither supplies family-aware paired method uncertainty. Every audited GO/EC result reuses proteins across scored pairs. [GO/EC audit](#).

The independently admitted recovered replay is reported separately from these historical method rows. Its profile branch had SwissTrees F1 differences of -0.003706 before and -0.003716 after sequence-based refinement (raw0-to1 units). The corresponding adjusted intervals were [-0.020824,0.003397] and [-0.020829,0.003347]. Across all four fixed stage contrasts and three metrics, every adjusted interval included zero; this establishes neither a benefit nor equivalence. Only CASP and GH14 changed under profile-branch processing (one family F1 gain, one loss, 16 ties), while sequence-based refinement changed only NOX (one loss, 17 ties). These are exhaustive descriptive family effects, not selected biological success stories or mechanistic attribution. The profile branch includes downstream graph/assignment responses; sequence refinement is not phylogenetic reconciliation. [Stage results](#), [paired intervals](#).

On the original-release inputs, OrthoMCL final-group scoring completed successfully, with all six endpoint tasks and consolidation exiting zero. Its secondary mean was 0.705177. The earlier pre-clustering diagnostic mean, 0.724414, is not the final-group assessment and is excluded from the principal comparison. The run contained 53 sequence-specific BLAST failures among 976,504 proteins; all 53 were absent from final groups. Direct-reference exposure was absent for VGNC, SwissTrees, TreeFam-A, and EC, but present for four experimentally annotated GO proteins; 46 failed proteins had nonempty FAS features. These observations do not bound indirect clustering effects or establish a negligible score impact. This historical baseline is retained with failures disclosed; neither its scores nor its failure count substitute for the corrected-release run. [Failure-impact audit](#).

Conserved-Family Recovery Remains A Weakness

Three Kingdoms BUSCO-reference F1 was 0.872133 for satellite_v2 and 0.826309 for high sensitivity, compared with 0.988582 for full OrthoFinder, 0.989451 for its sequence checkpoint, and 0.991276 for the contemporary matched-input SonicParanoid run (precision0.993443, recall0.989119; 7272TP,48FP,80FN). Historical Sonic F1=0.990794 remains a diagnostic result with a known input mismatch. All values measure reference-gene group co-membership, not genome-wide orthology or native pair prediction accuracy. The checkpoint's small advantage over full OrthoFinder on this restricted target does not establish that phylogenetic inference is generally harmful. Mechanistic explanations require error tracing and controlled ablations. [Recounted comparison with run/input labels](#).

Historical Components Provide Preliminary Evidence

Verified historical OrthoBench partitions gave F1 66.279051% after multipass grouping, 69.763388% with cluster refinement, 66.825390% with profile expansion alone, and 70.358998% with both. The descriptive increment from profile expansion after refinement was +0.595610 percentage points. The final replay partition was byte-identical to the corrected historical fresh output. These are historical component observations without paired intervals, not a complete current-source factorial or HMM-free control. [Historical audit](#).

Corrected Simulations Do Not Establish An OrthoHMM Advantage

In the fixed-length stress panel, high sensitivity completed all 70 datasets and satellite_v2 completed 64. Six satellite_v2 failures persisted because species-tree inference lacked connected single-copy taxon coverage. No full OrthoFinder or parent-gated checkpoint output passed native admission, so this panel provides no admitted paired comparison with OrthoFinder. Native normalization diagnostics reproduced degenerate length fitting and nonfinite scores; competitor failure is not evidence of an OrthoHMM accuracy advantage. [Fixed interpretation](#), [native failure audit](#).

In the variable-length panel, high sensitivity, satellite_v2, and full OrthoFinder passed admission on 70, 67, and 65 datasets, respectively. Full OrthoFinder led every paired condition-mean F1 comparison against either OrthoHMM mode. Adjusted intervals excluded zero below OrthoFinder for all seven high-sensitivity contrasts and four satellite_v2 contrasts; satellite_v2 turnover, missing-data, and uneven-sampling intervals included zero. Baseline satellite_v2 and full OrthoFinder mean F1 were 99.46% and 99.94%. Satellite_v2 paired deficits were 11.74 percentage points under divergence (five paired seeds) and 12.28 points under combined divergence and turnover (eight seeds). Lower recall was the principal observed deficit, but its causal processing stage has not yet been established. [Variable interpretation](#).

Three variable-panel satellite_v2 failures again involved species-tree coverage. Five divergent OrthoFinder runs had nonfinite graph weights. Reconstruction of native normalization across all ten divergent seeds reproduced seven nonfinite within-species matrices in exactly those five failed seeds. Each affected fit had two non-self hits at one length product; the rank-deficient fit produced extreme intercepts and scale-factor overflow. The proteomes themselves had heterogeneous lengths. This local failure does not imply that every rank-deficient fit fails, and no competitor repair or post-hoc exclusion of additional finite runs was applied. [Normalization audit](#).

All corrected OrthoHMM runs constructed profiles, but neither simulation panel added profile-expansion edges. These panels therefore do not demonstrate an accuracy contribution from multi-sequence profile expansion, although the initial HMM-based search remains present. They are not HMM-free controls. Synthetic sequence evolution, limited seed counts, conditional comparisons, and absent realistic domain architecture constrain extrapolation to proteomes.

The variable-length figure shows admitted-run counts beside paired F1 effects and nominal/adjusted intervals. The fixed-length figure explicitly shows the absence of admitted comparator contrasts. Both are generated from the corrected machine-readable results without imputing failures; PDF/SVG versions and source-hash manifests accompany the PNGs.

Stronger Tree Perturbations Reduce Simulation Accuracy

All 210 supplied-tree OrthoHMM runs passed native admission, compared with 195 of 210 OrthoFinder runs. The 15 OrthoFinder failures were the same five divergent datasets under each supplied tree, rejected for nonfinite upstream graph weights. Supplying a tree did not repair that problem. The three original OrthoHMM inferred-tree failures completed under all supplied topologies; their missing inferred accuracy cannot be recovered from this observation. The complete analysis contains 537 scores and 23 failures across 560 arms.

Generating-minus-inferred mean F1 changes ranged 0.000 to 0.217 percentage points for OrthoHMM and -0.005 to 0.160 for OrthoFinder. Every adjusted interval for these comparisons included zero. NNI2-minus-generating F1 and recall intervals excluded zero below it in four OrthoHMM conditions (turnover, divergent turnover, uneven taxa and taxon-count control) and two OrthoFinder conditions (uneven taxa and taxon-count control). These 12 endpoints were the only adjusted intervals excluding zero; no precision interval did. This establishes a bounded sensitivity observation, not comparative superiority, biological equivalence, or robustness to arbitrary species-tree errors.

Of 420 upstream-artifact contrasts, 400 were equivalent, 18 unavailable and two different. Both differences involved OrthoFinder divergent seed 20261106, with changed OG0000004 alignments and raw gene trees. They remain in the analysis and preclude strict tree-only causal attribution for the affected comparisons. The comparison requires joint success: OrthoFinder uses five divergent seeds and ten elsewhere; OrthoHMM generating-minus-inferred uses nine divergent and eight divergent-turnover seeds, with ten for other comparisons. Failure-conditioned estimates may be biased. Full records and limitations.

Simulation sensitivity to supplied species trees

70 datasets | 560 planned method/tree outcomes | 23 failures retained | 126 exploratory endpoints

OrthoHMM satellite_v2 OrthoFinder full 3.1.5

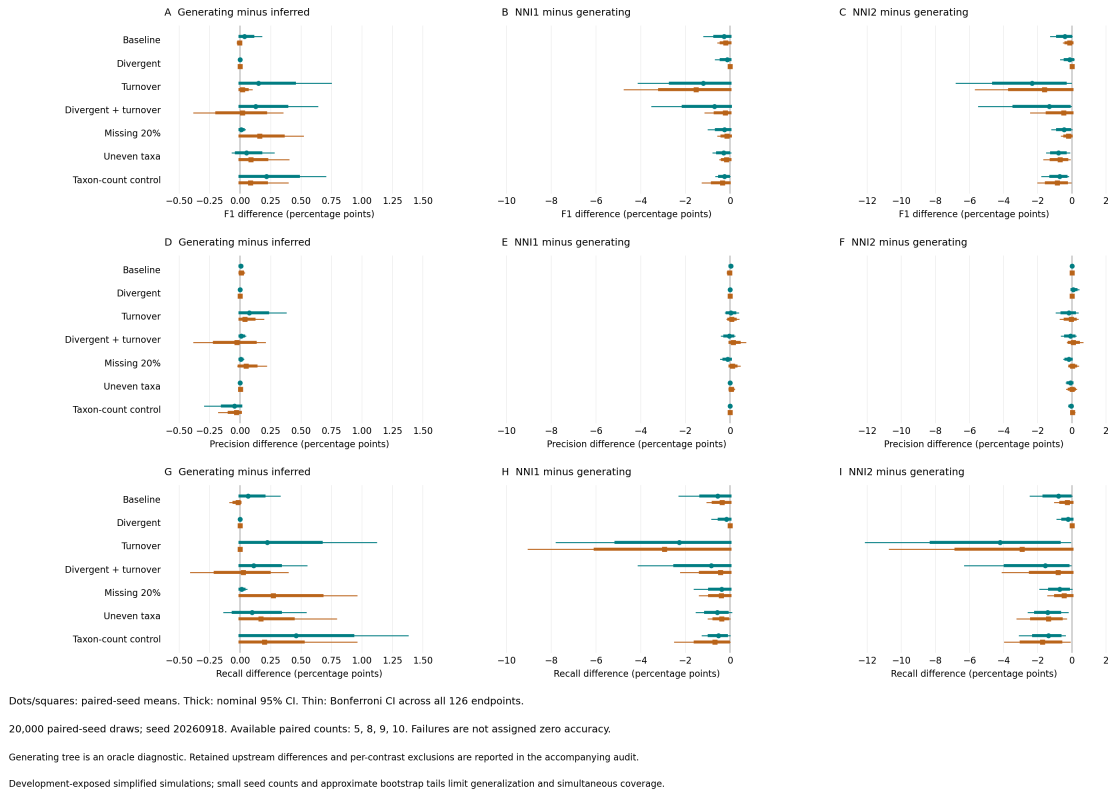


Figure: all126 exploratory F1, precision and recall endpoints across seven conditions, two methods and three tree contrasts. Points are paired-seed mean differences; thick intervals are nominal95% and thin intervals use Bonferroni adjustment over126 endpoints. Axes share scales down each contrast column. No best-tree selection or failed-score imputation was performed. The figure is generated from the complete verified summary with source/output hashes; PDF and SVG versions accompany the PNG.

Controlled Components Show A Candidate-Precision Tradeoff

The completed eight-cell OrthoBench factorial reproduced full-pipeline F1 74.106074%. Reconciliation improved F1 by 2.942-6.899 percentage points across four matched settings; all four multiplicity-adjusted intervals excluded zero above it. Broader candidates increased recall and reduced precision, with adjusted intervals excluding zero in all four settings. Their observed F1 effect changed from about -3.1 points without reconciliation to +0.697/+0.795 with reconciliation, but every adjusted candidate-expansion F1 interval included zero. Profile expansion added 0.568-0.704 observed F1 points; all adjusted profile-expansion intervals included zero. These contrasts do not remove the initial HMM search and do not establish a total HMM advantage.

All cells passed output/provenance gates and official-score crosschecks. Four batch failures caused by postflight working-directory-dependent package discovery were retained and separately recovered; native inference succeeded. Shared-node cached reconciliation costs are not end-to-end timings. The factorial remains development-exposed and supplies no new direct OrthoFinder comparison. [Results and interpretation.](#)

The original-release QfO factorial likewise completed all eight cells. Its prespecified SwissTrees family bootstrap used 100,000 shared draws and adjusted 42 endpoints. All 14 adjusted F1 intervals included zero. The four reconciliation contrasts increased precision by 30.02-34.02 percentage points and reduced recall by 4.56-6.81 points, with adjusted intervals excluding zero in both directions. Their observed F1 gains of 10.40-13.12 points did not exclude zero after adjustment. All six adjusted C-by-R interaction intervals included zero. This comparison changes native pair inference versus group-derived prediction semantics, not only group splitting. Only 18 development-exposed families were resampled; inference for other QfO endpoints remains unfinished. The completed corrected-release factorial is reported separately below. [Full results and limitations](#), [generated 42-endpoint table](#), [six-endpoint cell scores and prediction counts](#).

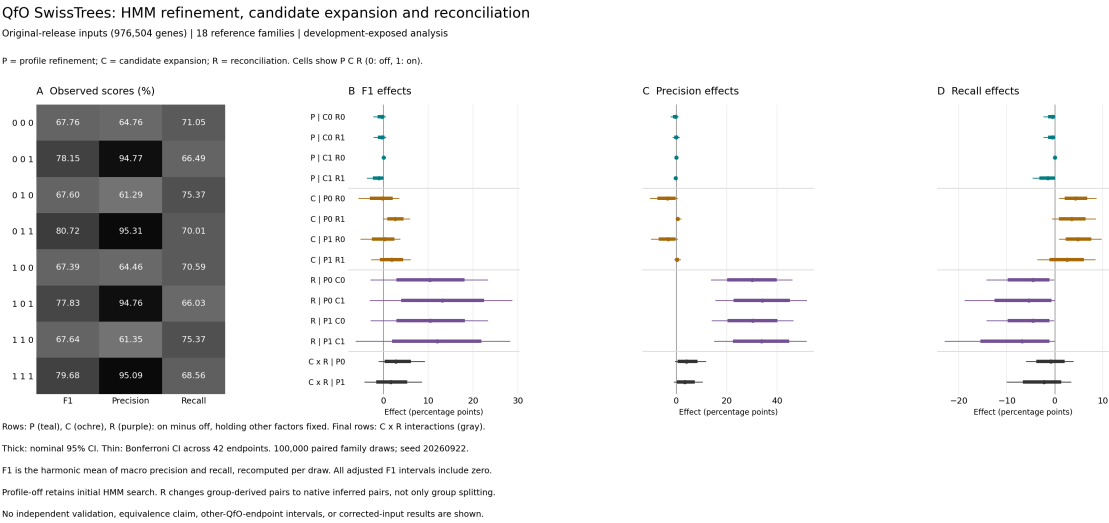


Figure: original-release SwissTrees cell scores and all 42 prespecified conditional-effect and interaction endpoints. Thick intervals are nominal 95% paired family intervals; thin intervals retain Bonferroni adjustment. Effects are shown in percentage points, including C-by-R differences of differences. This is not the corrected-input rerun or a comparison against OrthoFinder. The [figure manifest](#) binds the generated PNG/PDF/SVG files to their exact source result and plotter.

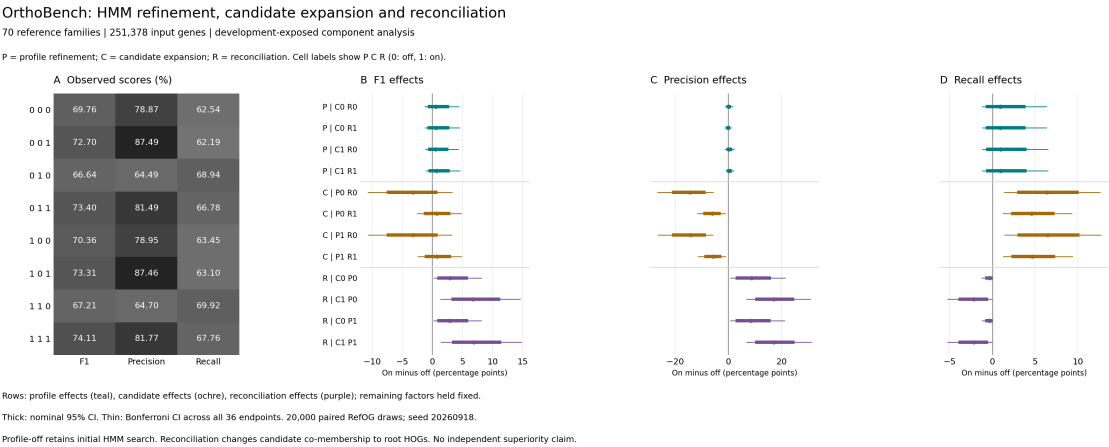


Figure: all eight cell scores and 36 conditional-effect endpoints. Thick intervals are nominal 95% paired RefOG bootstrap intervals; thin intervals retain Bonferroni adjustment across all 36 endpoints. Each effect holds the other two factors fixed. Profile-

off retains initial HMM search; this figure does not compare an HMM-free method or establish independent superiority.

An exploratory unconstrained replay used the same expanded candidate families and reconciliation rules but omitted membership constraints. Native output validation and official-score crosschecks passed. Its F1 was 72.614510%, precision 76.092405%, and recall 69.440641%, compared with 74.106074%, 81.770454%, and 67.755336% under constraints. Unconstrained-minus-constrained F1 was -1.492 percentage points (three-endpoint Bonferroni interval [-5.873, 0.931]); precision decreased 5.678 points ([-14.307, -0.794]) and recall increased 1.685 points ([0.013, 3.804]). Family F1 improved in 5, tied in 55, and declined in 10 RefOGs. These development-exposed results describe a precision-recall tradeoff, not a demonstrated F1 improvement or statistically established F1 loss. They do not justify removing constraints as a new default. [Diagnostic results](#).

Sequence-Search Replacement Control

With profile expansion, candidate expansion and reconciliation disabled, the frozen HMM-search baseline scored 69.763388% F1, 78.868592% precision and 62.542944% recall. DIAMOND all-hit replacement scored 65.762183/55.036231/ 81.680884%; the post-search top100 diagnostic scored 66.484803/55.897861/ 82.019046%. Six-endpoint adjusted F1 intervals for control-minus-HMM were [-14.834, 6.688] and [-14.424, 7.844] percentage points. Both include zero; these results do not establish an HMM-specific F1 advantage. Precision was lower and recall higher for both controls, with adjusted intervals excluding zero. Each control improved family F1 in 39, tied in 10, and declined in 21 RefOGs; the aggregate is not a mean of those family scores.

Hit sets were not sensitivity-matched. HMM/all-hit/top100 nonself counts were 18,235,373/99,848,167/48,740,683. Queries lacking cross-species hits numbered 33,351/24,526/24,526. DIAMOND all-hit recovery overlapped 65.12% of HMM hits, while only 11.89% of its nonself hits were shared with HMM. This asymmetry is not ground-truth sensitivity. The historical HMM cache omits self hits, whereas DIAMOND retains 250,980; raw no-hit query counts are not commensurate. All three final partitions cover 251,378 input genes, including singletons. The same normalization formula and E-value cutoff do not equate calibration, candidate filtering or computational effort. [Full control results](#).

[Sequence-search control figure](#) shows all observed scores and all six paired effects on a common percentage-point axis. Thin intervals retain the six-endpoint correction; thick intervals are nominal 95% intervals. Search sensitivity/calibration are not matched, and this control disables subsequent profile expansion and phylogenetic inference.

Supplied-Tree Perturbations Had Small Observed Aggregate Effects

Six prespecified rooted topology perturbations were evaluated against an unchanged supplied-tree control, with candidate groups, membership constraints, raw gene trees and reconciliation rules held fixed. The control exactly reproduced the inferred baseline partition and 74.106074% F1. Three variants had rooted clade distance 2 and three distance 4. Their F1 ranged from 73.744102% to 74.270946%, or -0.361972 to +0.164872 percentage points relative to control. Every one of the 18 Bonferroni-adjusted F1/precision/recall intervals included zero. Family F1 tied in 62-69 of 70 RefOGs per perturbation, although complete predicted partitions changed. All runs preserved 251,378 genes and passed native-provenance checks and official-score crosschecks.

These observations describe this limited, development-exposed topology panel; they do not establish equivalence, robustness to arbitrary species-tree error, or improved tree estimation. Intervals remain compatible with larger effects than the observed differences, and the variants are not posterior draws. No variant was selected to change the method. Recorded reconciliation costs reuse raw gene trees and are not end-to-end efficiency measurements. Full scores, uncertainty and coverage.

OrthoBench: sensitivity to supplied species-tree topology

70 reference families | fixed candidates and reconciliation rules | raw gene trees reused

Unchanged supplied control reproduces the inferred baseline. Six prespecified topology perturbations; no best-tree selection.

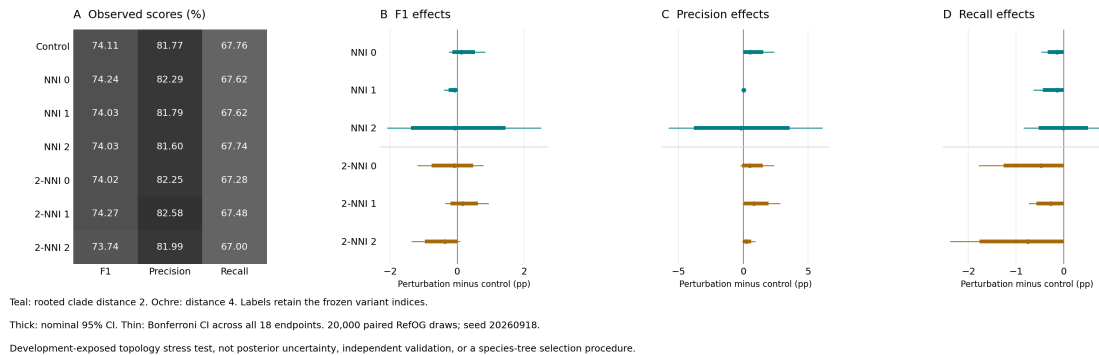


Figure: control and six fixed topology perturbations, ordered by the frozen variant indices. Thick intervals are nominal 95% paired RefOG intervals; thin intervals retain Bonferroni adjustment over all 18 endpoints using 20,000 paired resamples. Gene-tree checkpoints are shared; rooting/reconciliation is recomputed against each supplied species tree.

Exploratory Feature Strata Show Precision-Recall Tradeoffs

Fourteen feature strata were frozen before joining the new features to method errors. They describe reference-family size, copy number, canonical alignment identity, relative sequence length and global composition, not independently verified duplication history, fragments or domain architecture. All 70 RefOGs remain represented in each dimension. Twelve families have missing identity because at least one aligned pair lacks comparable canonical positions.

Both OrthoHMM configurations were compared with full OrthoFinder 3.1.5 for F1, precision and recall, retaining 84 endpoints in multiplicity adjustment. Eleven bins had at least five families, yielding 66 interval-bearing endpoints; one single-family composition bin was descriptive only and two empty bins were explicitly nonestimable. None of the 22 adjusted F1 intervals excluded zero. Nine adjusted precision/recall intervals excluded zero: two precision advantages and seven recall deficits for OrthoHMM across overlapping strata. These are not independent replications or tests of differences between strata.

In the 40-family short-relative stratum, high-sensitivity recall was 18.901 percentage points below OrthoFinder (adjusted interval [-36.475, -5.208]); phylogenetic-mode recall was 14.833 points lower ([-34.240, -0.370]). In the six large families, observed F1 differences were -16.469 and -11.616 points, but both adjusted F1 intervals included zero. These observations do not prove a fragmentation mechanism or justify outcome-selected parameter changes.

All three native-format parsers preserved 251,378 input genes, and fresh full-reference sufficient statistics and official-score crosschecks reproduced the frozen overall estimates. Stratified estimates restrict those original per-family counts without changing the reference or low-certainty conventions. Percentile intervals are approximate; the 84-endpoint adjusted tails contain only about six of 20,000 resamples each. All effects and limitations.

OrthoBench: stratified accuracy tradeoffs

OrthoHMM minus full OrthoFinder 3.1.5 | all 14 prespecified strata | 70 reference families

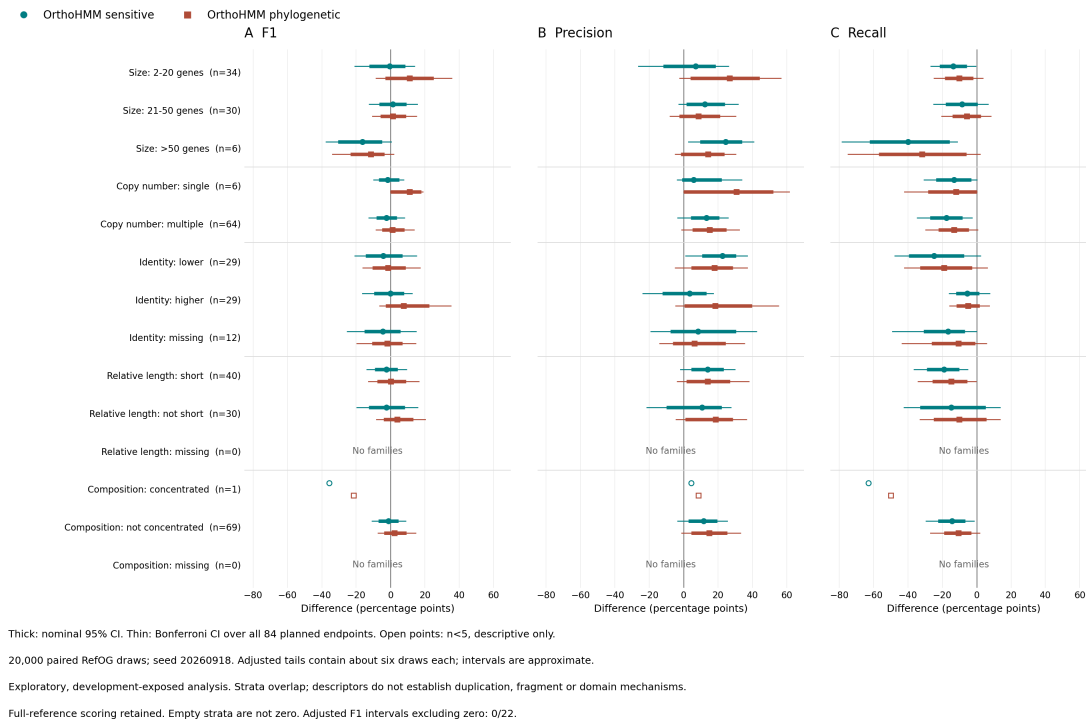


Figure: both OrthoHMM configurations minus full OrthoFinder3.1.5 across all 14 prespecified strata, with identical horizontal scales. Filled points show 66 interval-bearing endpoints: thick lines are nominal 95% paired intervals and thin lines retain the 84-endpoint Bonferroni adjustment. Six open points are descriptive estimates from a single-family stratum; 12 endpoints in empty strata remain explicitly nonestimable. Family counts are shown for every row. Strata overlap, and differences between their effect estimates are not interaction tests. Scores and points are checked against full-reference sufficient statistics; the figure does not add new inference runs or independent biological evidence.

Retained Checkpoints Localize Membership Changes

All 70 OrthoBench families were traced across six retained checkpoints, with 40,733 pair rows checked against native partitions. Both assignments of the one shared reference gene were preserved. The 8,440 logged candidate merges reconstructed the candidate partition; 168 events touched reference genes. These checks establish consistency of the retained evidence, not biological correctness. Profile expansion starts from unrefined multipass clusters; profile-off refinement is a separate branch, not its input.

Comparing matched refined endpoints, the profile-on branch gained 836 and lost 160 within-family pairs. Candidate expansion added 4,272; final tree, reconciliation and constraint processing removed 1,575. These raw co-membership counts include within-

species pairs and low-certainty assignments and are not official benchmark recall. All six hash-selected illustrations were retained, including neutral examples and candidate gains subsequently partly removed. Trace interpretation and full-family evidence. Rejected-search candidates, added profile-edge identities and per-family tree/constraint mechanisms still require inspection; no biological mechanism or new default is inferred from these membership changes alone.

Reconstruction of the 250 reference-incident candidate families distinguished root-lineage extraction from subsequent satellite membership constraints. For 114 reconciled and 136 bypassed candidates, saved node calls and all 193 local constraints exactly reproduced native final groups. Of 1,575 lost within-reference pairs, 134 were separated by root-lineage rules and 1,441 by subsequent constraints, affecting four and six families respectively (one family in both sets). Among the frozen illustrations, losses in RefOG014 and RefOG021 occurred at the constraint step. These are execution-order descriptions, not official recall, independent tree validation or a claim that constraints should be removed; the unconstrained control showed a precision cost without established F1 improvement. Full reconciliation trace and limitations.

Limited Parameter Sensitivity

The prespecified OrthoBench neighborhood changed CPM resolution, candidate minimum normalized score, or candidate minimum margin by 20% one at a time, retaining all other settings and the full inferred-tree pipeline. CPM variants recomputed their own HMM-refined seeds and candidates; threshold variants reused the verified baseline seeds. Exact-input raw-tree reuse was allowed, but supplied species trees were not substituted. All seven configurations retained 251,378 input genes, passed native admission, and matched the official OrthoBench scorer.

Control F1 was 74.106074%. F1 values were 74.586147% and 71.463468% for CPM0.08 and 0.12, 74.008570% and 74.110826% for minimum normalized scores 0.024 and 0.036, and 74.973144% and 73.274559% for minimum margins 1.2 and 1.8. Each of the six F1 contrasts had a Bonferroni-adjusted interval including zero, using 20,000 paired RefOG draws and all 18 planned F1/precision/recall endpoints. The only adjusted interval excluding zero was the recall decrease at CPM0.12: -2.010 percentage points, adjusted interval [-6.291, -0.148]. Minimum normalized score changes affected only one or three families' F1 values and left aggregate recall unchanged; this is a local observation, not general invariance.

The largest observed F1 increase, 0.867 points at minimum margin 1.2, had an adjusted interval [-1.362, 4.316]. It does not justify default promotion or an OrthoFinder superiority claim. This post-development panel is not independent confirmation, and intervals including zero do not establish equivalence. Shared-node phylogeny timings are incremental and omit upstream replay/candidate costs. Complete scores and paired intervals. Parameter sensitivity figure.

The corrected QfO threshold panel subsequently completed native inference, conversion, scoring and independent score validation for all four normalized score and margin variants. On the 18 SwissTrees families, control F1 was 83.3513%. Both normalized-score variants had identical family statistics to the control; their zero empirical contrast intervals describe this observed reference panel, not equivalence on unobserved families or identical whole-proteome predictions. Minimum margins 1.2 and 1.8 yielded F1 values of 82.8391% and 83.5648%, respectively. Their differences from control were -0.5122

percentage points (adjusted interval [-4.1173, 1.3523]) and +0.2134 points ([-0.9285, 1.9571]); each contrast had one family win, sixteen ties and one loss. Precision and recall adjusted intervals also included zero.

These intervals recomputed the native statistic in 100,000 paired family draws using the frozen seed and all 18 planned endpoints for multiplicity, including the CPM contrasts. The subsequent low-CPM result had SwissTrees F1 79.1014%, a difference of -4.2499 percentage points from control (adjusted interval [-21.9395, 0.7887]). Three families improved, thirteen tied, and two worsened; VATB had the largest family-level decline. A post-hoc retained-stage trace located its separation before phylogeny: low-CPM refinement changed two groups containing the 28 reference genes into 28 separate groups, including 27 singletons, and candidate expansion did not reunite them. The control retained two groups. The affected low-CPM group contained 276 genes and up to 12 copies from one species, satisfying the frozen broad-panel copy-split rule (at least 150 genes, 10 within-species copies, and 50 dataset species). All 276 members were singletons after refinement. This supports the copy-number guard as the explanation for this case, but is a post-hoc predicate/output check rather than a dynamic branch trace or evidence for changing defaults. Precision and recall intervals also included zero. High-CPM replay failed with SIGSEGV. Its checkpoint continuation completed, but independent validation and a subsequent allocator-debug refinement diagnostic both failed during post-write partition readback. Written partitions were byte-identical and covered the full gene universe, but this did not establish safe execution or authorize an accuracy result. The native cause remains unresolved. One subsequent prespecified refinement-only run under GDB completed normally with the same partition and a valid completion report; it did not reproduce the crash or provide a crash backtrace. This negative reproduction under instrumentation does not repair the failed admission or establish memory safety. No missing result was imputed. The full QfO parameter panel therefore remains incomplete. These findings do not justify changing frozen defaults and are development-exposed SwissTrees evidence, not a joint uncertainty analysis of all QfO metrics or independent confirmation. [Threshold results, provenance and reproduction](#). [Updated low-CPM uncertainty and provenance](#). [High-CPM validation failure evidence](#). [One-shot GDB diagnostic](#). [Post-hoc VATB stage trace](#).

Limitations And Unfinished Analyses

SwissTrees Input Coverage

A [frozen numeric-identity audit](#) found 14 SwissTrees reference proteins without mapped accessions in the frozen factorial FASTAs. All belong to the retained *Xenopus tropicalis* annotation resource. Their 577 incident reference relations have identical native labels across all eight retained tools: 181 false negatives and 396 true negatives, with no true or false positives. [The relation audit](#) preserves the full reference denominator and original scores. This common component does not establish ranking invariance under removal, historical input parity, or the cause of the input/reference resource mismatch. Those provenance questions and any separately declared sensitivity analysis remain outstanding; missing proteins are not relabeled as algorithmic fragment errors.

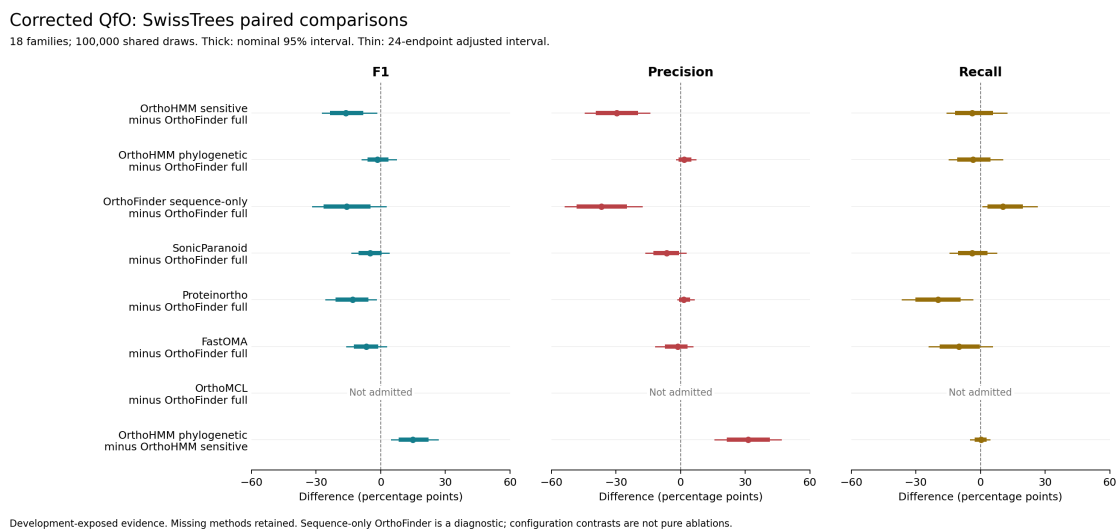
The subsequent [whole-input mapping audit](#) localizes a larger original-release mismatch to *Xenopus*: 990 input accessions are unmapped and 8,623 reference numeric identities lack an original input accession. All 78 canonical inputs match the original archive. EBI provides a separate corrected 2020 release for this proteome. Subsequent [archive comparison](#) and [native sequence/staging audits](#) establish compatibility with the retained

scorer: all 984,137 reference identities are uniquely represented, including the 14 previously missing SwissTrees proteins. Only the Xenopus canonical proteome differs from the original archive. Of the corrected sequences, 983,959 match the native reference exactly and 178 differ only by B/O/U/Z-to-X representation; no unexplained difference remains. The 78 canonical FASTAs were staged without residue normalization, with all identifiers, species ownership, sequence counts and hashes checked independently.

This resolves the observed input/reference mismatch, not its effects on method rankings. Original-input QfO scores and uncertainty estimates remain explicitly release-limited. The separately frozen corrected-release protocol requires fresh affected inference for all eight comparator rows and eight factorial cells, with separate prediction identifiers and fresh scoring. Corrected inference is in progress. Proteinortho is the first independently admitted corrected row: GO 0.486336, EC 0.963168, VGNC F1 0.954896, SwissTrees F1 0.718111, TreeFam-A F1 0.643187 and FAS 0.813595, with 4,695,385 mapped native pairs. Its project-defined secondary mean is 0.763216. SonicParanoid was subsequently admitted with GO 0.454385, EC 0.872761, VGNC F1 0.982794, SwissTrees F1 0.798459, TreeFam-A F1 0.771956 and FAS 0.736680, with 15,248,739 mapped native pairs and secondary mean 0.769506. A fresh frozen admission reproduced its complete report byte-for-byte. These two rows show different point-estimate trade-offs, not a uniform advantage. The partial corrected table does not yet complete the eight-method comparison; paired SwissTrees uncertainty for its admitted rows is now reconstructed separately below. The admitted high-sensitivity OrthoHMM row now adds GO 0.472271, EC 0.932038, VGNC F1 0.665933, SwissTrees F1 0.685498, TreeFam-A F1 0.605008 and FAS 0.776585 (secondary mean 0.689555), from 9,032,719 mapped group-derived pairs. This is p1_c0_r0, not the p0_c0_r0 initial-HMM ablation. The checked replay establishes its partition equivalence to the corrected native run. The final phylogenetic OrthoHMM row (p1_c1_r1) is also admitted: GO 0.490349, EC 0.965650, VGNC F1 0.901690, SwissTrees F1 0.833513, TreeFam-A F1 0.614864 and FAS 0.762993 (secondary mean 0.761510), from 5,959,560 mapped native phylogenetic pairs with zero mapping losses. SwissTrees precision/recall are 0.955177/0.739341, versus TreeFam-A 0.959110/0.452464. The table now contains seven admitted methods; OrthoMCL remains missing. Full OrthoFinder 3.1.5 adds GO 0.469548, EC 0.936130, VGNC F1 0.988546, SwissTrees F1 0.848413, TreeFam-A F1 0.791918 and FAS 0.691422 (secondary mean 0.787663), from 14,215,382 native phylogenetic pairs. Its pre-phylogenetic sequence-only MCL checkpoint adds GO 0.407043, EC 0.753642, VGNC F1 0.142755, SwissTrees F1 0.690515, TreeFam-A F1 0.734407 and FAS 0.561753 (secondary mean 0.548352), from 163,277,439 group-derived clique pairs. Both conversions have zero mapping losses, but their output semantics differ. The sequence-only checkpoint is a diagnostic, not native pairwise orthology output. Full OrthoFinder has higher point estimates on all three F1 endpoints than phylogenetic OrthoHMM, while OrthoHMM has higher GO/EC similarity and FAS. Full OrthoFinder SwissTrees precision/recall are 0.937853/0.774548, and TreeFam-A precision/recall are 0.924527/0.692578. Relative to OrthoHMM, this is lower precision and higher recall on both reference-tree tests, not a uniform advantage. FastOMA's corrected native pairs are now admitted: GO 0.436894, EC 0.897168, VGNC F1 0.950096, SwissTrees F1 0.780219, TreeFam-A F1 0.657902 and FAS 0.654366 (secondary mean 0.729441), from 15,008,180 native pairs with zero mapping losses. SwissTrees precision/recall are 0.923601/0.675373; TreeFam-A precision/recall are 0.882105/0.524572. FastOMA used a supplied corrected OrthoFinder species tree, not independent species-tree inference. Two failed native task attempts were retained and explicitly audited against their successful retries; their scientific commands were identical. This is accuracy evidence, not controlled runtime evidence. OrthoMCL's corrected comparison remains unfinished. Its interrupted search is being recovered through validated replay batches and an ordered merge; the downstream BPO, native-group, pair-conversion and scoring workflow is implemented and frozen, but this is not

evidence that those production stages have completed. The corrected OrthoMCL scores remain unavailable. [Recovery workflow and validation status](#). No paired superiority over competitors is inferred from this partial table. [Method identity and export checks](#). The prespecified [corrected comparator uncertainty analysis](#) initially reconstructed raw counts for the six methods then admitted. It retained all eight contrasts and the full 24-endpoint correction, using 100,000 shared family resamples with seed 20260920. Phylogenetic OrthoHMM minus full OrthoFinder F1 is -0.014900 (adjusted interval [-0.087078, 0.072090]); its precision and recall intervals also include zero. Neither superiority nor equivalence is established. Relative to high sensitivity, phylogenetic OrthoHMM increases F1 by 0.148015 ([0.049791, 0.266015]) and precision by 0.314178 ([0.158952, 0.466178]); recall's adjusted interval includes zero. This configuration contrast includes candidate expansion, not only phylogeny. Full OrthoFinder has higher F1 than high-sensitivity OrthoHMM and Proteinortho under the adjustment. The sequence-only checkpoint has lower precision and higher recall than full OrthoFinder, while its adjusted F1 interval includes zero. SonicParanoid's three adjusted intervals include zero. In that retained six-method analysis, FastOMA and OrthoMCL contrasts were unestimable, not zero. Independent arithmetic reproduction matches all 18 available endpoints within 1e-12.

The [seven-method update](#) reuses the same frozen executor, 18 families, 100,000 shared resamples and full 24-endpoint adjustment. FastOMA minus full OrthoFinder has F1 difference -0.068194 (adjusted interval [-0.158907, 0.026407]), precision difference -0.014252 ([-0.115839, 0.056552]) and recall difference -0.099175 ([-0.239207, 0.054470]). All three adjusted intervals include zero; neither superiority nor equivalence is established. Family-specific F1 differences favor FastOMA in one family, tie in one, and favor OrthoFinder in 16. These counts are descriptive, not a substitute for the macro-statistic bootstrap. [Arithmetic reproduction](#) matches all 21 available endpoints within 1e-12. OrthoMCL remains missing.



This figure retains every planned contrast. Thick bars are nominal 95% intervals and thin bars are adjusted across all 24 endpoints. Values are percentage-point differences; the linked data retain raw units. These conditional, development-exposed family intervals do not quantify uncertainty on other QfO endpoints or the secondary mean, and cannot establish a release effect by comparison with historical significance labels.

The prespecified [corrected composition-strata analysis](#) is now complete for the three primary configurations. Nine families fall in each input-only entropy bin. With 100,000 paired resamples and adjustment across 27 endpoints, phylogenetic OrthoHMM minus

full OrthoFinder F1 is -0.028710 (adjusted interval [-0.154364, 0.147961]) in the lower-entropy bin and -0.006102 ([-0.089129, 0.068670]) in the higher-entropy bin. All adjusted precision/recall contrasts for that comparison also include zero. Compared with high sensitivity, the phylogenetic configuration improves precision in both bins and F1 in the higher-entropy bin under the adjustment. All adjusted higher-minus-lower interactions include zero. Thus these data do not establish equivalence to OrthoFinder or a composition-specific causal mechanism. The configuration contrast is not a pure phylogeny ablation. All 27 endpoints appear in the linked figure, with an independent numerical reproduction from the admitted counts. The [all-method descriptive tables](#) now include seven admitted methods and explicitly missing OrthoMCL across all frozen primary and secondary bins. In seven families containing a member shorter than half the family median, F1 was 68.679% for high-sensitivity OrthoHMM, 83.365% for phylogenetic OrthoHMM and 85.546% for full OrthoFinder; the corresponding values in the other eleven families were 68.408%, 83.313% and 84.360%. These secondary summaries have no new inferential claims and do not diagnose fragmentation. Empty concentrated-composition and explicit-fragment bins remain missing, not zero. FastOMA's supplied tree and sequence-only OrthoFinder's group-clique semantics remain explicit.

An additional [exploratory identity-stratum analysis](#) froze the alignment and split protocol before joining the features to outcomes. Corrected reference sequences from all 18 families were aligned with MAFFT 7.525, with membership and ungapped residues checked against the inputs. A separate scalar calculation reproduced all family mean canonical pair identities. The median family identity, 0.429122, divides the panel into nine lower-identity and nine higher-identity families. All seven admitted methods have lower descriptive F1 in the lower-identity bin. Phylogenetic OrthoHMM scores 73.973% versus full OrthoFinder's 76.705% in that bin, and 91.914% versus 92.600% in the higher-identity bin. High-sensitivity OrthoHMM scores 64.023% and 72.435%, respectively. The complete table retains missing OrthoMCL and empty bins. No new subgroup intervals or significance claims are made. Sequence identity is alignment-dependent, not calibrated evolutionary divergence; correlated differences in taxa, domains, size and composition prevent a causal interpretation.

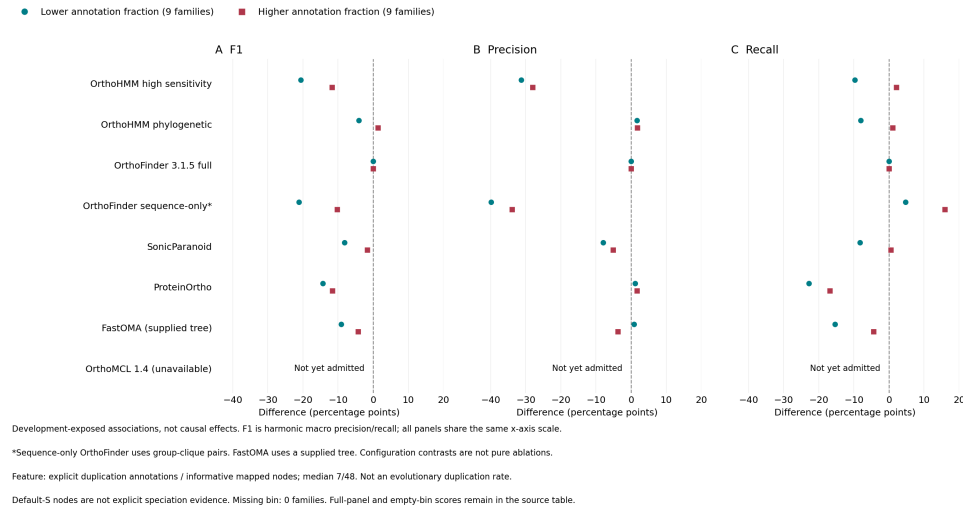
A separate [historical fragment-annotation analysis](#) matched all 563 reference proteins to sequence-identical UniSave entries; 549 matched the baseline-release interval and 14 required later records. Eleven proteins carried fragment/incomplete-sequence annotations across five families. Phylogenetic OrthoHMM and full OrthoFinder scored 83.827% and 87.926% F1 in these families, versus 83.143% and 83.498% in the thirteen all-matched unflagged families. Unflagged is not proven complete. Treating later-version annotations as missing leaves eleven unflagged families and changes that contrast to 84.562% versus 83.539%; the two excluded families score 73.586% versus 83.062%. Both views are retained because the sign change depends on family composition. These descriptive, development-exposed associations do not establish a causal fragmentation effect or superiority; no new subgroup inference is claimed.

A [prespecified duplication-annotation analysis](#) used the original trees embedded in the retained reference, with all 563 mapped memberships and 10,765 reference relationships reconstructed exactly. Separate Python and Darwin traversals agreed on 128 explicitly annotated duplication nodes among 546 informative mapped nodes. The other 418 nodes used the reference's default-S convention, not explicit speciation evidence. The median annotation fraction divided the panel into nine lower and nine higher families. In the [complete descriptive table](#), phylogenetic OrthoHMM scored 86.378% F1 versus full OrthoFinder's 90.453% in the lower stratum, and 80.309% versus 78.991% in the higher stratum. In the latter, precision/recall were 93.092%/70.613% for OrthoHMM and

91.320%/69.596% for OrthoFinder. High-sensitivity OrthoHMM scored 69.876% and 67.218% F1 in the lower and higher strata. All seven admitted methods had lower absolute F1 in the higher stratum. The small positive OrthoHMM contrast there does not reverse the full-panel ordering (83.351% versus 84.841%) or establish subgroup superiority. This annotation fraction is reference-derived and related to the benchmark's own pair labels; alias collisions, default-S nodes, family composition and development exposure limit interpretation. No new subgroup intervals, causal claims or duplication-rate interpretation are made. Missing OrthoMCL remains NA.

QfO SwissTrees: mapped-tree duplication annotations

Descriptive differences versus full OrthoFinder 3.1.5; no subgroup intervals or significance tests.

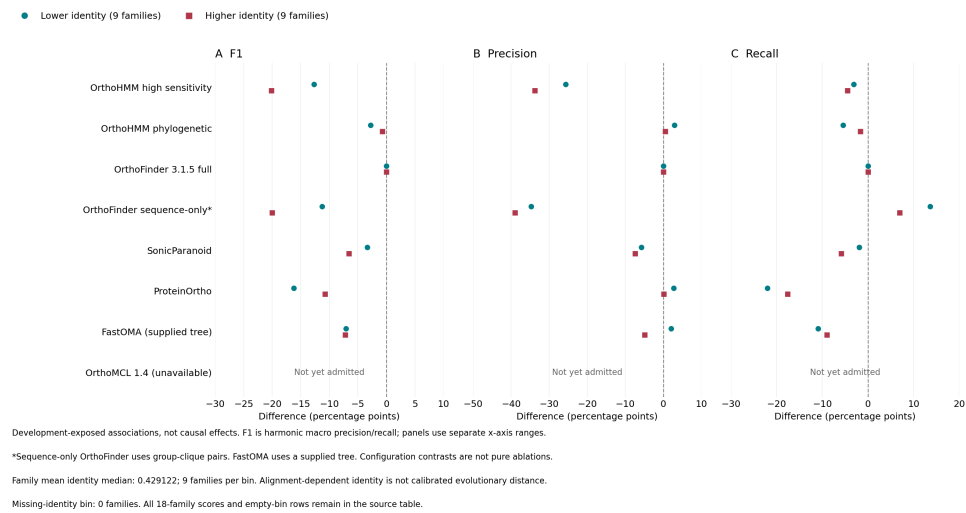


Descriptive duplication-annotation differences in F1, precision and recall versus full OrthoFinder.

Supplementary Figure: Duplication Annotations. Points show differences for the nine lower and nine higher annotation-fraction families, with a common horizontal scale across metrics. These are observed differences, not confidence intervals or significance tests. The feature is derived from the reference trees, not an evolutionary rate or an independent explanation of benchmark errors. Full-panel scores and the empty missing bin remain in the source table; unavailable OrthoMCL is not plotted as zero. [Figure data and provenance](#) retain all 48 endpoint records, including six unavailable values.

Ofo SwissTrees: Sequence identity

Descriptive differences versus full OrthoFinder 3.1.5; no subgroup confidence intervals or significance tests.

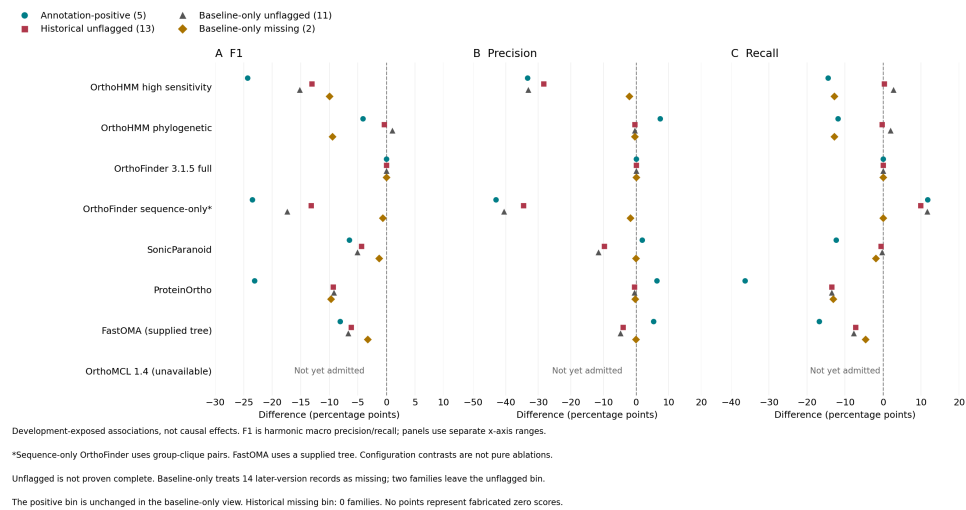


Descriptive identity-stratified differences in F1, precision and recall versus full OrthoFinder.

Supplementary Figure: Sequence Identity. Points show observed differences in percentage points for the nine lower-identity and nine higher-identity families. No subgroup intervals or significance tests are added. Separate x-axis ranges are used for each metric. Missing OrthoMCL remains unavailable; full OrthoFinder is the zero-difference reference. Configuration differences are not isolated phylogeny effects, and identity is not calibrated distance.

Ofo SwissTrees: Historical fragment annotations

Descriptive differences versus full OrthoFinder 3.1.5; no subgroup confidence intervals or significance tests.



Descriptive historical fragment-annotation differences, including baseline-only sensitivity views.

Supplementary Figure: Fragment Annotations. Observed differences versus full OrthoFinder retain the positive (five families), historical unflagged (thirteen), baseline-only unflagged (eleven) and baseline-only missing (two) views. The positive family set is unchanged by baseline-only restriction. Unflagged does not establish completeness; later-version records become missing in the baseline-only view. No causal or inferential claim follows from a subgroup sign change. [Figure data and provenance](#) link both figures to the pinned descriptive tables and all 144 displayed endpoint records, including unavailable values.

Old groups, trees, scores and confidence intervals cannot be relabeled as corrected results. The previously measured 53 legacy BLAST query failures also belong to the original run; failures in the corrected OrthoMCL search must be measured independently. Recovery of missing inputs does not turn this development-exposed benchmark into independent validation.

A [sequence-content follow-up](#) found 974,363 exact byte matches among 975,514 mapped original inputs. All 178 same-length differences are solely B/O/U/Z-to-X representation changes, accounting for every non-Xenopus discrepancy. The remaining 973 length-different pairs are all Xenopus. This separates observed residue representation from the input-release issue without normalizing the frozen inputs or assuming all inference methods treat nonstandard residues equivalently.

Corrected QfO Replay Matches Native Grouping

The corrected-input high-sensitivity cached replay passed independent admission after all four native clustering boundaries completed successfully. Its final partition matched the corrected native result exactly: 391,908 groups covering 984,137 proteins, with no native-only or replay-only groups. The admission workflow independently reread the partitions and checked 419 provenance records, including the frozen runtime and scheduler identity. Retained intermediate outputs contained 308,697 multipass groups, 394,328 refined multipass groups, 306,051 strict-profile groups and 391,908 refined strict-profile groups. Group counts alone are not accuracy endpoints. [Independent admission and evidence](#).

This agreement supports reuse of the corrected intermediate outputs for the prespecified ablations. It neither resolves the older historical-input replay discrepancy nor establishes general determinism. Candidate preparation was admitted before the factorial assessments; conversion and scoring have since completed for all eight cells, as summarized below. The 50:28 replay elapsed time is incremental shared-host execution, not a matched end-to-end runtime result.

The corrected R-off cells p0_c0_r0, p0_c1_r0, p1_c0_r0 and p1_c1_r0 scored SwissTrees F1 of 0.689184, 0.685709, 0.685498 and 0.686087, respectively. Their group-derived predictions retained 9,009,082, 11,734,021, 9,032,719 and 11,755,521 cross-species pairs with zero mapping loss. These R-off cells alone do not establish expansion/reconciliation interactions or authorize a default change. [All six endpoints and admission provenance](#).

Native inference and pair conversion have also completed for p0_c1_r1, retaining all 984,137 genes and 5,977,100 native phylogenetic pairs with zero mapping loss; its accuracy assessment is now independently admitted. An exploratory comparison with p0_c0_r1 found that the inferred species trees differ: both have the same 78 species and 76 nontrivial rooted clades, but share only 38 clades (rooted symmetric difference 76). The runs used 18 versus 25 families for species-tree estimation. This contrast therefore includes downstream tree estimation, not candidate expansion under a fixed realized tree. Neither inferred tree is treated as truth, and the topology difference does not establish a cause of an accuracy change. [Native evidence and exploratory comparison](#).

First Corrected Reconciliation Contrast

Independent assessment admitted p0_c0_r1 with 5,113,820 native phylogenetic pairs and zero mapping losses. Compared with the matched p0_c0_r0 ablation, reconciliation increased precision and reduced recall in all three reference-orthology endpoints. These

comparisons retain initial HMM search but disable multi-sequence profile expansion and candidate-family expansion; neither row is the complete satellite_v2 publication configuration.

Endpoint	Reconciliation off	Reconciliation on
GO similarity	0.472119	0.490260
EC similarity	0.932114	0.967702
VGNC F1	0.666834	0.898185
SwissTrees F1	0.689184	0.789574
TreeFam-A F1	0.605404	0.602508
FAS	0.777125	0.785212

SwissTrees precision increased from 0.643940 to 0.949152 while recall fell from 0.741266 to 0.675932. TreeFam-A precision increased from 0.792245 to 0.956682, but recall fell from 0.489873 to 0.439719, leaving slightly lower F1. Thus a precision increase did not imply uniform F1 improvement. The project-defined secondary mean increased from 0.690463 to 0.755573; it is not a separate primary endpoint or evidence of general superiority.

All eight corrected factorial cells are now scored. Their [machine-generated table and manifest](#) retain all corrected-release admissions. R-off pairs are group-derived cliques; R-on pairs are native phylogenetic predictions, not root-HOG clique expansions. Pair counts describe prediction volume, not protein coverage. The prespecified full-factorial SwissTrees uncertainty analysis is complete, as reported below. Historical-input intervals were not reused, and no default was selected from these development-exposed point estimates. [Independent score admission and precision-recall details](#).

Corrected Profile Refinement With Reconciliation

With candidate expansion disabled and inferred reconciliation enabled, multi-sequence profile refinement changes SwissTrees F1 from 0.789574 to 0.786372. The corresponding p1_c0_r1 scores are GO 0.490229, EC 0.968271, VGNC F1 0.897844, TreeFam-A F1 0.602432 and FAS 0.784576; the secondary mean is 0.754954. Five of six point estimates decline slightly and EC increases. All 5113180 native pairs map without loss. These end-to-end point estimates do not establish a significant effect or direct mechanism; P-off still uses initial HMM search. All four corrected profile-refinement F1 intervals include zero after adjustment. [Validated scores and provenance](#).

Corrected Candidate Expansion With Reconciliation

With profile expansion disabled and reconciliation enabled, candidate expansion increased observed SwissTrees F1 from 0.789574 to 0.836354, precision from 0.949152 to 0.955781 and recall from 0.675932 to 0.743456. VGNC F1 increased from 0.898185 to 0.901626 and TreeFam-A F1 from 0.602508 to 0.616864. Other endpoints decreased: GO similarity from 0.490260 to 0.490035, EC similarity from 0.967702 to 0.965435 and FAS from 0.785212 to 0.762823. The secondary mean increased from 0.755573 to 0.762189. These mixed point estimates do not establish significance, a factorial interaction or superiority of the complete publication configuration. The species tree also changes between these cells, as described above; the

contrast is end-to-end, not fixed-tree isolation. The completed paired analysis below evaluates the interaction separately; no historical interval is reused. Independently admitted scores and precision-recall trade-offs.

Complete Corrected Factorial Uncertainty

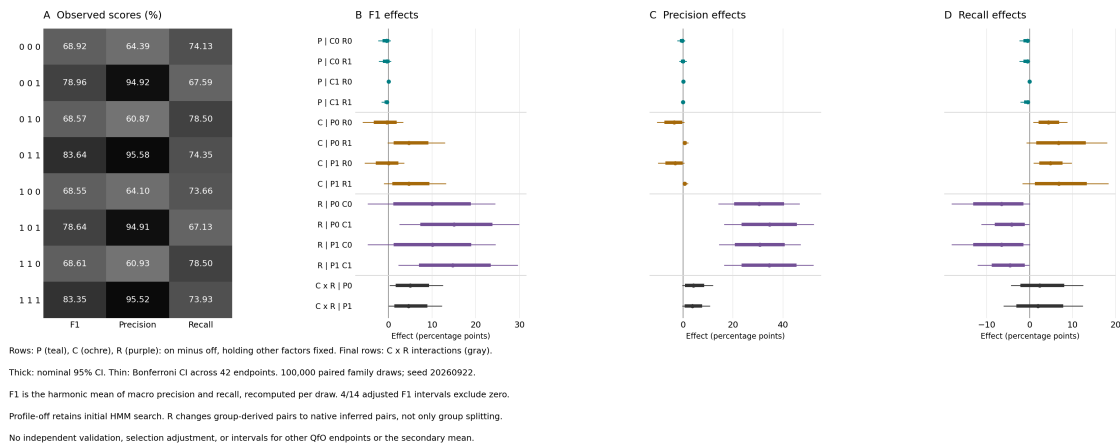
The full p1_c1_r1 configuration has SwissTrees F1 0.833513, precision 0.955177 and recall 0.739341. Its six-metric secondary mean is 0.761510. All 5,959,560 native pairs map without loss. Using 100,000 shared draws over 18 families (seed 20260922), with the actual benchmark statistic recomputed per draw and Bonferroni adjustment across 42 endpoints, reconciliation at C1 increases F1 by 0.150644 [0.026173, 0.298879] at P0 and 0.147426 [0.024122, 0.295846] at P1. Candidate-expansion by reconciliation F1 interactions are 0.050254 [0.003650, 0.124556] and 0.046552 [0.001825, 0.121851], respectively. The other ten F1 intervals include zero. All four reconciliation precision intervals are positive, while all four recall intervals are negative.

These conditional, development-exposed results support an end-to-end interaction, not a fixed-tree mechanism or independent generalization. Only 18 families are available, and adjustment does not account for prior development selection. No corrected full-OrthoFinder superiority is established. Fresh frozen admission, count-audit and bootstrap executions reproduced the corresponding JSON outputs byte-for-byte. Complete scores, intervals, family effects and provenance.

QfO SwissTrees: HMM refinement, candidate expansion and reconciliation

Corrected-release inputs (984,137 genes) | 18 reference families | development-exposed analysis

P = profile refinement; C = candidate expansion; R = reconciliation. Cells show P C R (0: off, 1: on).



The corrected-release figure shows all eight cells and all 42 effects; thick intervals are nominal 95% and thin intervals use the prespecified 42-endpoint Bonferroni adjustment. Effects are percentage points, whereas the numerical differences above are raw 0-to-1 units. Profile-off retains the initial HMM search. This figure is separate from the original-release factorial figure and does not replace its historical evidence. Figure provenance and reproduction.

Corrected QfO Sequence-Search Control

Independent validation admitted the DIAMOND all-hit control with frozen OrthoHMM downstream grouping:11,300,151cross-species clique pairs survived reference mapping without loss. Endpoint scores were GO similarity0.479327, EC similarity0.874662, VGNC F1=0.602031, SwissTrees F1=0.628092, TreeFam-A F1=0.576306 and FAS0.715448. The project-defined secondary mean was0.645978. The independently

admitted top100diagnostic retained11,285,357mapped pairs without loss and scored0.479791/0.874979/0.604172/0.629189/0.573917/0.716734 on the same six endpoints, with secondary mean0.646464. The same-input corrected HMM control is now admitted (SwissTrees F1=0.689184). In the prespecified 100,000-replicate paired family bootstrap with six-endpoint adjustment, all-hit minus HMM F1 was -0.061092 [-0.188023, 0.028017] and top100 minus HMM F1 was -0.059995 [-0.186513, 0.029034]. Both precision differences favored HMM (all-hit -0.082110 [-0.181914, -0.001387]; top100 -0.082123 [-0.181927, -0.001404]); both recall intervals included zero. Thus these development-exposed controls support a bounded precision finding, not an established F1 improvement. Equal cutoffs do not establish matched sensitivity or effort. Source-bound counts and uncertainty. Both admitted controls; top100provenance.

Sequence-control uncertainty figure. The score table shows percentages. The three panels show all six DIAMOND minus initial-HMM effects in percentage points, with a common symmetric axis and zero reference. Thick lines are nominal paired 95% intervals; thin lines are Bonferroni-adjusted intervals over the six endpoints. Both F1 and recall intervals cross zero. Precision intervals exclude zero but remain conditional on these 18 development-exposed families. The render manifest binds the figure to the admitted result and renderer.

The completed label-free search diagnostic compares initial search checkpoints on the same984137proteins and78species. HMM search retained90,687,327directed hits, DIAMOND all-hits593,510,904 and the top100control321,164,891. Queries with at least one cross-species hit comprised74.44%,76.78%and76.78%of the input, respectively. Among non-self directed hits,48,603,572were shared between HMM and DIAMOND all-hits, representing54.1825%of HMM hits and8.2027%of DIAMOND hits. The HMM hit set is therefore not a subset of DIAMOND's. These are output-set differences, not calibrated sensitivity or independent orthology accuracy estimates.

Search diagnostic figure. Panel A shows directed-hit counts including self-hits; panel B shows the fraction of all input queries with at least one cross-species hit, not reference-relative recall. Both axes start at zero. The HMM checkpoint precedes multi-sequence profile expansion and cannot measure full-pipeline recovery. Top100 limits reporting per query and target species after search; it is not a global100-hit limit or an HMM prefilter emulation. No confidence intervals are attached to these complete-set descriptive counts. The figure is generated from the source-bound summary, with render provenance.

Remaining Requirements

No universal superiority, arbitrary-dataset generalization, or controlled speedup is established. Frozen YGOB evaluation is complete and supports novel-taxon transfer, not family-disjoint confirmation. Homolog-family overlap remains substantial even after taxon exclusions. The proposed HMM contribution to F1 is not established by the completed exploratory OrthoBench or corrected QfO sequence-search controls. Corrected SwissTrees precision differences favor initial HMM under the prespecified adjustment, but F1 and recall intervals include zero; independent confirmation remains required.

The corrected-release QfO factorial supports a bounded SwissTrees finding: reconciliation at expanded candidates improves F1 at both profile settings, and both candidate-expansion-by-reconciliation F1 interaction intervals exclude zero under the frozen 42-endpoint adjustment. The interactions are +0.050254 [0.003650, 0.124556] with profile refinement off and +0.046552 [0.001825, 0.121851] with it on, in raw 0-to-1 units. All four profile-refinement F1 intervals include zero. Reconciliation increases precision and reduces recall at each setting. This is conditional evidence from 18

development-exposed families, not independent confirmation or a mechanistic isolation of reconciliation: candidate expansion can alter tree estimation, and reconciliation changes pair-output semantics. The earlier original-release QfO interaction intervals included zero; both analyses are retained, not conflated. No superiority over corrected full OrthoFinder is established. [Complete corrected factorial](#).

Corrected-input comparator evaluation and better-matched search controls remain outstanding. Corrected multi-seed simulations and their generating-tree/NNI controls are complete but do not establish an OrthoHMM advantage or profile-expansion benefit. Tree perturbations show bounded sensitivity, not arbitrary-error robustness. QfO and independently annotated error strata, mechanistic tracing, more realistic evolutionary conditions and the QfO parameter panel, matched resource scaling, and deeper biological application mechanism tracing remain required.

TreeFam-A Pooled Count Audit

The separate [TreeFam-A count audit](#) reproduced all four recovered-stage native endpoints from 79,320 raw reference relations. This challenge stores one pooled case, not separately labeled families. Of 11,140 mapped reference proteins, 11,130 occur in relations; ten are isolated reference members and do not represent failed predictions. Native confusion counts equal raw one-direction counts divided by two plus one prior per pooled category. Appropriate family-level uncertainty remains unresolved until original source-family mappings are recovered and validated. No independent-pair bootstrap or degenerate single-case interval is substituted.

Prespecified Whole-Genome-Duplicate Application

Experimental duplicate-pair evidence comes from [Kuzmin et al. \(2020\)](#), with the exact retained table recorded in the prospective application protocol.

The frozen application retained all 240 experimental duplicate pairs, of which 239 had both anchors in the prepared inputs and 231 shared an unambiguous YGOB reference pillar. These *Saccharomyces* data are development-exposed, not an independent generalization test. Full native runs and their output identities were admitted before scoring. OrthoFinder was evaluated using its retained root HOG table with bijective restoration of original sequence identifiers; unassigned proteins were not supplemented with synthetic singleton groups.

High-sensitivity OrthoHMM separated 58/239 pairs, phylogenetic OrthoHMM 238/239, full OrthoFinder 236/239 and SonicParanoid 229/239. Requiring each separated anchor group to retain at least one non-*S. cerevisiae* reference homolog yielded 56/231, 193/231, 227/231 and 223/231 supported separations, respectively. Mean reference-homolog coverage was 99.149%, 82.338%, 98.413% and 98.773%. High coverage alone is insufficient: merged paralog groups can retain every reference homolog. Conversely, splitting without homolog support can inflate the unqualified separation rate.

Under the prespecified paired pillar bootstrap (20,000 replicates; seed 20260920; Bonferroni adjustment across 12 contrasts), phylogenetic OrthoHMM improved supported separation over high sensitivity by 59.307 percentage points (adjusted interval 49.784 to 68.398). However, it trailed full OrthoFinder by 14.719 points (difference interval -21.645 to -8.225) and lost 16.075 points of homolog coverage (-19.755 to

-12.496). Against SonicParanoid, the corresponding deficits were 12.987 points (-19.913 to -6.494) and 16.436 points (-20.101 to -12.973). Thus, the results do not support superior biological recovery by phylogenetic OrthoHMM.

These endpoints measure homolog-supported paralog separation, not correct cross-species assignment to ancestral duplicate copies. All exclusions, experimental strata, six prospectively selected examples and the diagnostic OrthoFinder MCL checkpoint remain in the [machine-readable report](#) and [complete pair table](#). The [application figure](#) uses the same 231-pair population for all panels. Native-membership recalculation with separate scoring arithmetic reproduced all 1,200 method/pair records, class summaries and six selected examples. A direct pair-resampling calculation reproduced all 12 intervals; this simplification is valid here because the 231 pairs occupy 231 distinct pillars. Native-format readers are shared, so this does not constitute an independently implemented parser validation.

The [six prespecified cases](#) include success, partial recovery, a support failure and a reference exclusion. YDR122W/YLR096W was merged by high sensitivity but separated with all six non-S. cerevisiae homologs retained by phylogenetic OrthoHMM, full OrthoFinder and SonicParanoid. For YER059W/YIL050W, all four methods separated the anchors and retained all six homologs. In contrast, phylogenetic OrthoHMM retained five of six homologs for each of YER132C/YGL197W and YBR147W/YOL092W, whereas OrthoFinder and SonicParanoid retained all six. In both cases the reference pillar intersected three OrthoHMM root HOGs rather than the two anchor groups. For YCL048W/YDR522C, phylogenetic OrthoHMM separated the anchors but retained only three of six homologs, all in one anchor group; the other anchor group had no reference-homolog support. Full OrthoFinder and SonicParanoid retained all six with support for both anchors. The sixth example, YLR284C/YOR180C, remains reference-excluded because its anchors map to different pillars. None was replaced after inspecting outcomes.

The [retained-artifact case trace](#) reconstructed seven incident candidate families, including two that bypassed tree inference. It required exact agreement between reconstructed final groups and admitted native membership. All five non-S. cerevisiae homologs outside the anchor groups in the three partial-recovery examples were present in their anchor candidates. They separated during root-lineage grouping, before the satellite constraints were applied, and remained assigned to other final HOGs. Candidate omission and subsequent constraint splitting therefore do not explain these five observed coverage losses. Two incident families had unsupported logged constraints, but those did not change these focal coverage outcomes.

This localizes the computational stage, not the underlying biological error. The trace reproduces recorded node calls rather than independently validating tree topology, rooting or duplication history. Node-table and family-checkpoint hashes are retrospective, although tree/checkpoint consistency and exact final membership were checked. Earlier search effects on candidate composition and gene-tree inference remain possible. The two OrthoHMM configurations differ in more than reconciliation, so their contrast is not an isolated phylogeny effect. No assertion that every extra group is erroneous follows from this homology-only reference.

QfO historical replay equivalence remains unresolved. A bounded capture verified identical initial RBNH graph arrays and gene order but a different first clustering partition than an earlier diagnostic. Thus the observed discrepancy precedes profile refinement; its specific cause is not yet established. No replay result is silently substituted for the historical baseline. [Diagnostic evidence](#). Three subsequent instrumented single-CPU workers with matching recorded native-library/runtime identities reproduced the

earlier diagnostic partition, but not the replay capture. A subsequent controlled affinity panel exposed within-configuration variability: the two one-CPU repeats differed (349,898 versus 349,950 groups), although their recorded sources, libraries, inputs, settings and affinity matched. Both 32-CPU repeats produced 349,898 groups. Independent file and complete-partition admission confirmed the discrepancy. Thus CPU availability alone does not explain the variation, and the earlier matching repeats do not establish general repeatability. The cause remains unresolved; no preferred partition is substituted for the historical baseline.

A subsequent native-boundary diagnostic stopped during its second worker before optimization: ordered native endpoint hashes differed from the saved graph despite matching vertex/edge counts and ordered weights. The first worker had matched the saved graph before and after optimization. Fresh saved-array hashing confirmed the expected endpoint fingerprint. This localizes a discrepancy to pre-optimizer graph handling in that instrumented run, but does not yet distinguish conversion, construction or observation effects, or establish the cause of all historical partition differences. The failed partial panel is preserved and not presented as a completed reproducibility experiment.

A second boundary diagnostic recorded six incorrect native edges despite an intact original constructor array. A completed construction-only follow-up alternated three original-int32 workers with three explicit-int64-copy workers. Five graphs matched the saved endpoint stream; one explicit-int64 worker had the same six endpoint mismatches. Both its original and converted arrays were intact. Native tuple and source/target access agreed on the observed endpoints, and edge lookup did not find the expected pairs. Independent admission checked 257 file records and reconstructed each worker's full reported native endpoint hash from the saved arrays plus its complete bounded mismatch witnesses. Explicit conversion alone is therefore not a sufficient fix in this experiment. The audit validates preserved observations, not an independent inspection of the historical live object; it does not establish a specific library defect, hardware cause, or the cause of all historical variability. No optimizer was invoked and no partition or accuracy score was produced in this follow-up. [Construction evidence and limitations](#).

A direct-construction follow-up completed six workers alternating minimal and frozen-worker imports. Four had the same six endpoint mismatches before weight assignment, two matched, and each worker's recorded differences were unchanged after assigning weights. Both modes contained mismatching workers with intact constructor arrays. Independent admission checked 278 file records and reconstructed all post-weight endpoint hashes. Pre-weight hashes are implied by bounded complete witnesses, not separately recorded full hashes. Neither weight assignment nor OrthoHMM/Leiden imports is necessary for this observed failure; its underlying cause remains unresolved. This is preserved-observation evidence, not independent live-object inspection or proof that every historical result is affected. [Direct-stage results](#).

A six-worker input-format comparison subsequently completed: all three streamed Python-integer-pair constructors matched the saved graph, while one of three NumPy constructors had the same six pre-weight endpoint mismatches. Independent admission checked 278 file records and the full post-weight graph fingerprints. This bounded result motivates integrity-gated follow-up, not a proven fix or general repeatability claim; input format also changes allocation behavior. No optimizer or accuracy score was produced in this comparison. [Input-format results](#).

Three subsequent integrity-gated initial-graph optimizer runs completed with byte-identical partitions of 349,898 groups covering all 976,504 genes. Each used Python-integer-pair construction and preserved all 24,148,515 ordered undirected edge endpoints and weights before and after unchanged CPM optimization. Independent admission

reconstructed saved hashes, checked 308 provenance records, and recomputed complete-coverage partition comparisons for every pair of runs. This supports bounded repeatability of the checked initial-graph path, not general determinism, a causal library diagnosis, or historical complete-pipeline equivalence. No accuracy score was used to select an output; full cached-stage validation remains required. [Checked-repeat results](#).

Historical timing and memory records differ in scope and accounting. Cached replays are incremental computations, not end-to-end timings; supplied trees have upstream costs. Unscheduled competing workloads were observed, so exclusive Slurm allocation alone cannot establish controlled CPU conditions. GNU-time maximum RSS and simultaneous process-tree RSS must not be pooled as if they were the same measurement.

A subsequent dedicated ARM DGX panel completed 27 sequential runs at 20 CPUs and 96 GiB per allocation. Retained metadata and host-monitor streams replayed successfully for all runs, as did the resource summaries from 63,445 observations. However, all host assessments remained inconclusive. The maximum observed persistent foreign load was 0.2112 CPU cores, below the fixed 0.25-core monitor threshold, but this does not bound unmatched or entirely unsampled work. Of 1,908 unmatched identity events, 1,710 were kworker-named; names were not treated as kernel authentication.

Process-read gaps were method-dependent: 838 affected samples for full OrthoFinder, 3,868 for phylogenetic OrthoHMM, and none for high-sensitivity OrthoHMM. These 4,706 samples contained 18,695 error events, mostly NoSuchProcess (18,627). One OrthoFinder run had a process-read error at its recorded aggregate-RSS maximum. Error-free sampled maxima in other runs do not exclude missed between-sample peaks. Consequently, sampled aggregate RSS does not support an unqualified cross-method memory ranking. Cgroup memory and GNU-time maximum process RSS retain their distinct scopes. All 27 native outputs subsequently passed input-universe, completion and method-specific output checks. The complete archived inventory comprises 1,273,699 files and 18,018,339,602 bytes, including frozen inputs. Per-run observations and three-repeat medians/ranges are retained as descriptive results, not controlled-workload speed comparisons. No fastest-repeat selection, speedup ratios, timing-based exclusions or automatic replacements were applied. These limitations remain despite successful output validation. [Native validation](#), [descriptive resource observations](#), [all 27 runs](#). The [descriptive resource figure](#) shows all repeats, median ticks and observed ranges for elapsed time, CPU use and cgroup peak memory. Ranges are not confidence intervals. Sampled aggregate RSS is deliberately not plotted; no speedup or complexity fit is implied. Host uncertainty is stated within the figure itself. [DGX observation-gap audit](#), [post-run validation status](#).

A separate prospective 18-task observation-overhead experiment subsequently finished. It compared periodic and boundary-only monitoring in three paired replicates for each principal method on the four-proteome input. Independent audit validated six task records; three measurement wrappers failed and nine tasks lacked the required detailed scheduler records because accounting summaries had been collected instead. Only one of nine planned pairs was available. Its 1.986% periodic/boundary wall-time difference does not establish the prespecified complete-panel overhead budget or an interference bound. No missing pair was imputed and no failed arm selectively replaced. [Complete audit and failures](#).

In two failed wrappers, retained snapshots identify transient apt-daily and NetworkManager-dispatcher cgroups as the reason for identity-check rejection; their effect on native runtime is unquantified. The third failed inside a snapshot and lacks the two inventories needed for that diagnosis. All three native exit records report success, distinguishing measurement failure from demonstrated inference failure. Subsequent

engineering changes automate scheduler-record capture and preserve partial failed-snapshot evidence, but do not retroactively recover these missing records or admit the panel.

An independent host-pressure audit checked 5,154 retained points and their source hashes. Seventeen observation windows enclosed native execution; task 6 covered only 10.038107 seconds before monitoring failed. Recorded host memory-stall totals were small (0-227 microseconds for the some category), while CPU and I/O pressure varied. Host-wide pressure includes native work and does not identify competing processes or establish isolation. No pressure threshold was selected after observing these outcomes. The original 27 runs remain descriptive; controlled speed and memory rankings are not established.

A subsequent pressure-enabled overhead panel retained all 18 detailed terminal records and validated 16 tasks, yielding seven of nine complete pairs. Three-pair median periodic/boundary changes were -0.710027% for high-sensitivity OrthoHMM and +0.077954% for satellite_v2; only one OrthoFinder pair was available. Negative differences represent variation, not negative measurement cost. The complete-panel overhead budget remained unestablished. Pressure-overhead audit.

Three prospective native diagnostics then retained two host-counter brackets around the same native CPU observations. All native products matched the prescribed prior outputs. The narrower bracket flagged 0/551 intervals for high sensitivity, 21/819 for satellite_v2 and 1/622 for full OrthoFinder; original wider-bracket flags were preserved. These non-atomic residuals do not identify foreign CPU activity. Native diagnostic audit.

A prespecified nine-trial control experiment compared steady work, repeated short-lived process creation and steady work with a known outside-target competitor, in three balanced blocks. Twenty native workers were pinned to the allocated CPUs for 20 seconds; independent witnesses checked scope, affinity, exits, CPU dose and at least 19.5 seconds of common work. All nine trials passed those checks. The narrow screen flagged 0/126 native-only intervals and 63/63 contended intervals, including all 57 fully enclosed common-work intervals. Churn increased within-block median residuals by 0.07845-0.08762 cores; contention increased them by 0.94738-0.94930 cores. Both increased native CPU pressure, so pressure alone was not specific evidence of outside work. Control audit, descriptive comparisons.

Post-outcome recomputation on fixed 5-, 10- and 30-sample endpoint grids reduced satellite_v2 narrow flags to one, zero and zero, respectively, and OrthoFinder flags to zero at each scale. Known sustained competitors remained detectable. Every final partial window and original interval flag was retained. Aggregation can obscure brief real interference as well as accounting noise; it therefore does not establish a cause or justify retrospective acceptance. The controls support bounded monitor behavior, not controlled speed rankings. Window-scale diagnostic.

The subsequent 18-task dual-collector overhead panel is complete: 16 tasks passed native-output/provenance and measurement-replay checks, while two periodic satellite_v2 measurements failed when transient service cgroups appeared during sampling. Seven of nine assigned pairs are available. Median signed periodic-versus-boundary wall-time differences were 0.6067% for high-sensitivity OrthoHMM and 1.0411% for full OrthoFinder. Each met its 5% median numerical budget, but the satellite_v2 median and complete-panel budget remain unestimable. All seven available pairs met the 60-second duration and 10% per-pair numerical gates and had matching canonical outputs. These within-method collector comparisons are not tool speed rankings. Twenty-three narrow interval flags remain in the seven validated periodic tasks;

failed observations were not dropped or selectively retried. No scientific timing inclusion rule has been adopted, and controlled comparative timing remains unestablished. [Completed audit](#), [machine-readable summary](#), [prospective protocol](#).

An aggregate-lineage collector subsequently completed all three native diagnostic workloads without the earlier sibling-inventory failure. Provenance, raw-measurement replay and native-output validation passed; canonical outputs matched the pinned same-method predecessors, with no temporal-order or boot-domain discrepancies. Original interval flags numbered 1, 56 and 0 for high-sensitivity OrthoHMM, satellite_v2 and full OrthoFinder, respectively; the corresponding narrow counts were 1, 7 and 0. These flags remain retained. Non-atomic ancestor-counter differences do not identify their cause or bound interference. Successful output reproduction therefore does not establish environmental validity or admit comparative timings. [Native lineage audit](#).

A complete new 18-task periodic-versus-boundary lineage overhead experiment was frozen with the same workloads, paired order and numerical budgets. All 18 tasks completed and validated, with all nine paired work/duration checks passing. Median signed periodic-versus-boundary wall changes were +0.085529%, -0.505169% and +1.614059% for high sensitivity, satellite_v2 and full OrthoFinder, respectively. All numerical overhead budgets passed. Negative values reflect variability, not acceleration by monitoring. The panel retained 143 original and 23 narrow interval flags; boundary-only observations explicitly lack interval coverage. Remaining CPU discrepancies and prospective scientific timing inclusion remain unresolved. No timing correction or scientific speed ranking follows. [Completed audit](#), [Protocol](#), [deployment and submission](#).

Exploratory inspection retained all 5,929 periodic intervals, including the 23 narrow flags. Flagged root-minus-system.slice CPU differences were positive, but non-atomic counters did not identify their cause. [Complete interval description](#). In a separate prespecified engineering control, three finite owned services each consumed approximately 0.75 process-CPU seconds and disappeared between the lineage reader's root and next ancestor reads. All three controls and their raw evidence replay passed. Full-span signed root-minus-observer responses were 0.809154, 0.814570 and 0.811985 CPU seconds; the first partial spans were negative and retained. This demonstrates bounded lifecycle handling and read-window skew, not the cause of native flags, attribution specificity, arbitrary-churn robustness or controlled comparative timing. [Read-crossing control and replay](#).

A subsequent 12-trial root/user-slice context panel stopped at its first user-contended condition when the user-manager session had already shut down. The initial idle, steady and churn trials replayed successfully, with zero original or narrow flags in their 63 total observation intervals; the failed condition and eight unrun conditions are retained. This incomplete panel does not establish sensitivity to the intended user-service load or scientific timing eligibility. [Failure and raw replay](#).

The separately identified held-session follow-up validated all 12 controls. All three user-contended trials exceeded the fixed five-CPU-second aggregate response criterion, with 57/57 common-work intervals flagged versus 0/171 idle/native-only intervals. An unrelated restarting user daemon remained present and is documented. This supports the tested monitor response under the recorded environment, not background-free timing, exact task attribution or context-collector overhead on native tools. [Complete follow-up and replay](#).

The subsequent native integration panel completed all three frozen four-proteome commands with replayable supplementary root context and canonical outputs equal to the prior lineage diagnostics. Across 1,976 intervals, 52 original and 11 narrow flags

remained; none were removed. The bounded waiting session was verified, but an unrelated failing user service scheduled 383 restarts during the job window. These results establish native collector integration, not background-free measurements, causal attribution, collector overhead or comparative timing eligibility. Native integration and retained flags.

A prospective paired panel subsequently validated all 18 native tasks and nine lineage-versus-root-context comparisons against the same-method native baseline. Median signed elapsed-time changes were -0.929631%, +1.342841% and -0.344162% for high sensitivity, satellite_v2 and full OrthoFinder. All pairs met the prespecified +10% limit and all complete method medians met +5%. All 344 original and 60 narrow flags across 11,846 intervals were retained. Fixed order and an unrelated restarting user service limit interpretation: these are incremental collector comparisons, not causal overhead estimates, method-speed rankings or scientific timing admission. Paired overhead results and provenance.

The retained non-CPU assessment found positive native-step I/O pressure in all 18 overhead runs and positive memory pressure in the six satellite_v2 and six full-OrthoFinder runs; the six high-sensitivity runs recorded zero memory pressure. All recorded memory-event counters, including OOM events, were zero. These are observation-window counters, not exact command-only stall times or evidence identifying outside interference. Neither zero native pressure nor absence of sampled foreign processes establishes isolation; requiring zero pressure could selectively exclude legitimate native work. Device-I/O attribution, GPU activity over the full run, thermal effects and memory-bandwidth interference remain uncharacterized. Non-CPU assessment and retained records.

A prospective replacement protocol preserves all 27 original method/size/repeat identities, scientific settings, nested input bytes and order, with fresh outputs and no automatic retries. Each intended allocation is exclusive, with 20 CPUs, 96 GiB, a 24-hour limit and a 23-hour-50-minute native timeout. The long-run amendment corrects the short diagnostic collector's incompatible timeout contract before launch; it does not alter scientific parameters or historical measurements. Native failures retain consumed resources, while infrastructure or policy failures pause further submissions. No fastest-repeat selection, pressure subtraction or observed-overhead correction is planned. Complete three-repeat summaries require three eligible, output-valid runs, with unavailable outcomes visible.

The replacement panel has not been executed. Short collector probes verified temporary suppression and restoration of the explicitly authorized failing user service, terminal scheduler handling, and relocated raw replay; these are infrastructure diagnostics, not method timings or environmental admission. The service's missing application remains unrepaired. On 23 September 2026, the user deferred further DGX work. Task binding, raw replay and outcome-classification tests do not establish whole-run environmental validity or native-output correctness. The actual service/workload policy, complete environmental observations and controlled replacement timings remain unmet. The original 27-run panel remains descriptive evidence only; local queued accuracy analyses are not replacements for controlled comparative timing.

SwissTrees Comparator Uncertainty

For all eight retained historical comparison outputs, raw SwissTrees counts covered the same 18 reference families and 10,765 labeled relations. Reconstructed native per-family precision and recall, with confusion counts $\text{raw}/2+1$, reproduced the retained macro

statistics. A committed follow-up protocol specified 100,000 shared family-bootstrap draws (PCG64 seed 20260920), seven contrasts against full OrthoFinder and one contrast between OrthoHMM configurations. Every replicate recomputed macro precision/recall and their harmonic mean. Percentile intervals were adjusted across all 24 contrast/metric endpoints.

All seven comparator-minus-full-OrthoFinder adjusted F1 intervals were negative. For phylogenetic OrthoHMM, the difference was -0.067184 (adjusted interval [-0.127608, -0.021232]); for high sensitivity it was -0.185310 [-0.285081, -0.091574]. Phylogenetic OrthoHMM had lower recall than full OrthoFinder; its precision difference interval included zero. Against high sensitivity, phylogenetic OrthoHMM had a positive F1 point difference of 0.118125, but its adjusted interval [-0.007676, 0.241518] included zero. Its precision increase of 0.304810 [0.150217, 0.455082] remained positive under this adjustment. All differences are in raw 0-to-1 units.

These are approximate conditional sensitivity estimates for 18 curated, development-exposed families, not independent or model-selection-adjusted confirmation. Disjoint represented genes do not remove shared evolutionary history or dependence from merged predictions. Inclusion of zero does not establish equivalence. The OrthoFinder checkpoint remains a diagnostic, FastOMA used a supplied tree, and the OrthoHMM configuration contrast is not a pure reconciliation ablation. The results do not quantify uncertainty for other QfO challenges or the six-metric secondary mean. [Full results and family contrasts](#). [The uncertainty figure](#) shows every prespecified contrast in percentage points, with nominal and adjusted intervals; the [figure note](#) provides its caption, validation scope and reproduction command.

Annotation-Defined SwissTrees Error Strata

An annotation-only inventory found exact accession matches for all 563 represented SwissTrees proteins in the retained QfO FAS resource. Five records had no Pfam hits, distinct from absent annotation. A committed follow-up protocol split the 18 families by median distinct Pfam types: 12 below two and six at least two. The secondary repeated-type split contained 15 and three families and was reported descriptively without inferential intervals. These annotations are external to predictions but are not independent validation of the FAS endpoint.

The planned within-bin family bootstrap used 100,000 draws (PCG64 seed 20260921), recomputed macro precision/recall and harmonic F1, and adjusted 27 endpoints covering three method contrasts, two primary bins and their interaction. Both OrthoHMM configurations had negative adjusted F1 differences versus full OrthoFinder in both primary strata. Phylogenetic-minus-sensitive OrthoHMM F1 was +0.185688 in the higher-type bin (adjusted interval [0.024605, 0.413865]) and +0.085765 in the lower-type bin [-0.074822, 0.222339]. All nine adjusted interaction intervals included zero; the F1 interaction for that configuration contrast was +0.099922 [-0.121894, 0.379765]. A within-bin interval excluding zero in only one bin does not establish that the effects differ between bins.

This retrospective comparison is not a causal domain-effect analysis. Few curated families, shared history, merged predictions and correlated differences in taxon sampling, divergence or family size limit interpretation. Pfam types and repeated hit instances are annotation features, not independent fragment, domain-loss or ancestral duplication labels. No bin or threshold was changed after observing stratified outcomes, and no

method was retuned. [All strata and contrasts](#). The [domain-stratified figure](#) shows all 27 endpoints, including the nine neutral adjusted interactions; its [caption and reproduction note](#) documents the fixed source and validation.

VGNC Prediction-Database Audit

For the four recovered sequence-refinement stages, a separate audit reconstructed native true-positive, false-positive and false-negative pair sets from the mapped prediction databases. Every category matched the retained raw output exactly, and precision, recall and harmonic F1 matched the admitted endpoints within $5e-8$. The native reference contains 23,934 asserted pairs and 36,986 proteins; 11 proteins have multiple family labels. Native final-label and accession-alias behavior was retained and verified against raw annotations.

The multipass, multipass-refined, strict-profile and strict-profile-refined stages respectively predicted 232,314, 40,460, 232,521 and 40,520 pairs among reference proteins. Of these, 88,316, 4,739, 87,865 and 4,798 were neither asserted true positives nor eligible native false positives. Thus the precision denominator does not include every predicted pair even within this reference subset. This is native benchmark scope, not a newly imposed exclusion. These checks validate scoring of these databases, not upstream conversion completeness, independent family resampling, or every publication competitor. [Reference and prediction audit](#).

A [dependency-structure audit](#) collapsed overlaps among 16,863 family labels, yielding 16,844 reference blocks. All scored true positives and false negatives lay within these blocks, whereas 120,804, 15,788, 121,468 and 15,804 false positives crossed blocks in the four stages, respectively; each stage had only two within-block false positives. Thus merging overlapping reference labels alone does not justify ordinary independent-family resampling of the native precision denominator. Connected components of prediction links also vary with the method and are not used as outcome-defined resampling units. These observations explain an unresolved dependence problem; they provide neither confidence intervals nor proof that valid paired inference is impossible. VGNC remains a point-estimate endpoint with an explicit uncertainty limitation.

An exploratory [fixed-table influence diagnostic](#) then deleted each reference block's incident scored rows separately. Across all 16,844 deletions, neither refinement's positive F1 difference nor the strict-profile stages' small negative F1 differences changed sign. The refinement differences ranged from +39.0496 to +39.2274 percentage points without strict profiles and +39.1141 to +39.2909 with strict profiles. These dependent deletion ranges are not confidence intervals. This checks single-block influence on historical scored tables only; it neither reruns native eligibility nor establishes generalization or comparator superiority.

Data And Code Availability

Historical installation manifests retain their original tool versions and must not be interpreted as secure installation recommendations. The original CPU-wheel lock pins pip and setuptools versions affected by 11 retained repository advisories. A separately tested [patched development-installation lock](#) and [local-wheel, smoke and advisory-range evidence](#) are available without altering frozen benchmark environments. The September 20 recheck finds no affected recorded versions in that patched installation for those 11

ranges. This does not establish a secure complete build chain, current installed-environment clearance, portable release or closed repository alerts; the historical lock remains evidence only, not for new installation.

A primary-source data-rights register distinguishes QfO deposit, UniProt, WGD, BUSCO, OrthoBench and YGOB terms. In particular, BUSCO software and dataset licenses differ. OrthoBench/YGOB redistribution and exact Three Kingdoms input notices remain unresolved. No repository-wide license or local archive is treated as blanket clearance for acquired material; final file-level and software-notice review is pending.

A relocatable figure-evidence bundle preserves 18 retained panels, their 61 outputs, original manifests and directly recorded dependencies from committed source revisions. Its standalone verifier passed after archive extraction outside the repository. This is a byte-identity and relocation check, not regeneration of all scientific results, native inference, or clearance of redistribution rights. The archive remains local and is not the complete publication deposit.

A corrected-QfO figure supplement separately preserves the corrected factorial, composition-stratified and comparator figures and their direct dependencies from commit 5132dda. Its 27-file export, including 12 outputs, passed isolated verification after extraction outside the repository. The earlier 11-file factorial export from 706fc03 is retained; historical plotter identities are not overwritten. This is evidence preservation, not statistical or native-inference regeneration. These archives have not been deposited, and redistribution review remains incomplete.

The corrected factorial statistical reproduction separately exported committed counts and numerical code outside the checkout. All eight cell estimates and all 42 paired endpoints reproduced exactly under isolated Python with the frozen 100,000 draws and seed. This does not repeat native inference, official scoring or raw-source admission.

A relocated reproduction workflow exports the SwissTrees comparator analysis from a fixed commit and reruns it in a fresh hash-pinned Python environment. Its numerical results and Markdown table reproduce exactly, and the figure workflow completes. This validates a bounded statistical workflow, not native inference or raw benchmark acquisition and scoring, and does not resolve third-party redistribution permissions.

The OrthoBench factorial relocation check also exported committed analysis inputs and code outside the checkout. All eight cells and the 20,000-draw paired analysis reproduced exactly from the 70-family sufficient statistics in an isolated Python invocation. This used the existing analysis virtual environment on the same host, not a fresh installation or independent native-inference/scoring reproduction.

Code and analysis milestones are tracked in the authorized repository. Machine-readable reports retain input, source, and output hashes where audited; large raw datasets and working outputs are not included in normal source commits. The figure bundle is generated from the comparison JSON and includes a provenance manifest. Absolute local paths are provenance, not portable download locations. A portable workflow bundle, complete dependency lock, raw-output inventory, redistribution-permission review, versioned release, and archival deposition remain unfinished. No archive accession or publication DOI has been assigned by this work.

See the [claim-to-evidence checklist](#) before strengthening conclusions or describing this package as publication-ready.